The Future of Labor: Automation and the Labor Share in the Second Machine Age*

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Abstract

We study the effect of modern automation on firm-level labor shares using a 2018 survey of 1,618 manufacturing firms in China. We exploit geographic and industry variation built into the design of subsidies for automation paid under a vast government industrialization program, "Made in China 2025," to construct an instrument for automation investment. We use a canonical CES framework of automation and develop a novel methodology to structurally estimate the elasticity of substitution between labor and automation capital among automating firms, which for our preferred specification is about 3.0. We then calibrate the model and find that automation accounts for a substantial part of the labor-share decline in our sample, even though automating firms produce only about a quarter of value added.

Key words: labor share, automation, labor demand, industrial robots.

JEL Classifications: D33, E25, O33, J23, J24, E24, O25.

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1 Introduction

The share of labor in national income has shrunk globally by about five percentage points since the 1980s, a large movement by historical standards (Karabarbounis and Neiman, 2014).[1] One hypothesis is that this decline is driven by a modern wave of automation: increasingly sophisticated "robots" (fueled by advances in artificial intelligence, machine learning and dexterous automation) now erode humans' advantage in both routine and, more recently, non-routine cognitive tasks.[2] Yet despite widespread popular concern,[3] direct causal evidence on how modern automation affects the labor share remains scarce. This paper provides such evidence. We estimate the causal effect of automation on the labor share of individual firms, recover the structural elasticity of substitution between automation capital and labor that drives it, and use a calibrated general-equilibrium model to gauge the aggregate implications.

Measuring the effect of modern automation on the labor share is challenging because firms do not automate at random. Those that invest in automation tend to be systematically different (larger, faster growing, more profitable, or more capital intensive to begin with), so a raw comparison of their labor shares reflects this selection rather than the technological effect of automation itself. Worse, the forces behind the selection are exactly the channels that rival explanations for the aggregate decline emphasize, so each could drive the labor share on its own.[4] Isolating the causal technological effect of automation therefore requires an identification strategy that shifts automation for reasons orthogonal to these forces. Such a strategy would reveal whether the link between automation and the labor share is indeed a technological one.

In this paper, we attack this problem by developing a novel approach that uses policy-induced

[1]See Dao et al. (2017) for updated evidence. See also Autor et al. (2003), Goos and Manning (2007) and Acemoglu and Autor (2011) for evidence on how automation is impacting jobs and job polarization.

[2]For example, according to Manyika et al. (2019), a quarter of labor hours is estimated to be lost to automation by 2030, with highly skilled occupations expected to be at stake as much as low skilled occupations. Similarly, Muro et al. (2019) estimate that "approximately 25% of U.S. employment will face high exposure to automation in the coming decades—with greater than 70% of current task content at risk of substitution." See also Frey and Osborne (2017). For a contrarian view, see the work by Arntz et al. (2016). While their estimates are considerably smaller, they are nonetheless sizable.

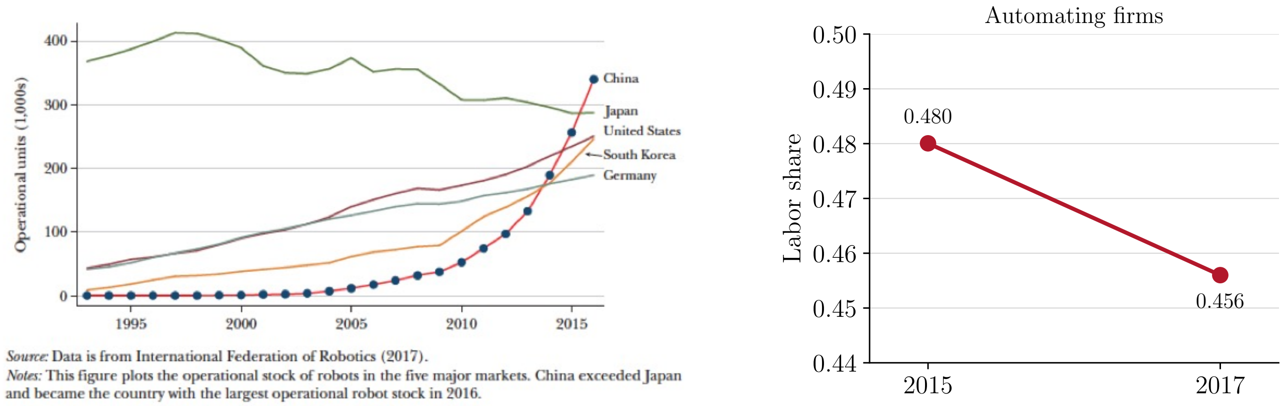
[3]See, for example, Brynjolfsson and McAfee (2014), Ford (2009) or Frey (2020).

[4]A firm hit by a favorable demand shock, for example, might expand and, when its demand elasticity varies with output, earn a higher markup, lowering its labor share even as it invests in automation to meet the rising demand; the resulting comovement of automation and the labor share is not causal. Offshoring of labor-intensive activities to lower-income countries can have a similar effect, as firms specialize in more capital-intensive goods. In a similar vein, Hubmer (2020) shows that changing consumption patterns may partly account for the decline in the labor share in the US.

local variation in the price of automation capital as an instrument for automation investment at the firm level. To implement our approach, we use new data from a survey of 1,618 Chinese manufacturing firms conducted by the Wuhan University Institute of Quality Development Strategy and designed in collaboration with a number of researchers from the Hong Kong University of Science and Technology, Stanford University and the Chinese Academy of Social Sciences.[5] Our data provides detailed information on firms' operations between 2015 and 2017, including the type of equipment they purchased and, most importantly, the subsidies that they received from the government for purchasing automation capital (industrial robots and numerically controlled machines). These subsidies were paid under an unprecedented government industrialization program, "Made in China 2025" (MIC, hereafter), and they were implemented by local municipalities. As a result, subsidy rates varied considerably across cities and industries. We exploit this variation in the effective price of automation capital to identify the causal impact of automation on the labor share. Figure 1 sets the stage for our analysis: panel (a) shows China's rapid ascent to the world's largest market for industrial robots, and panel (b) the concurrent decline in the labor share of automating firms in our sample.

To develop our empirical strategy, we extend a canonical model of automation (along the lines of Graetz and Michaels (2018) and Acemoglu and Restrepo (2018)) and lay out how, in that theory, after adding a large amount of firm- and industry-level heterogeneity in parameters and shocks, the presence of orthogonal variation in subsidies for automation capital can identify the causal impact of automation on the labor share. We state the assumptions under which this identification scheme is valid, and discuss its validity and limitations in the context of our analysis. We then show how to structurally estimate the key parameter responsible for that relationship in the model: the elasticity of substitution between labor and automation capital. We further use the model as a testing ground for the estimator itself: applied to data the model simulates from a known elasticity, our procedure recovers it, confirming that the identification is sound. While we use this analysis in the context of a particular dataset, our approach is general

[5]China is a good case to study the impact of automation as it has been one of the most aggressive adopters of robots in the world in the past decade. The staggering pace of robot adoption in China is evident from the fact that China's share in the global market for robots went from 3.7 percent in 2005 to around 30 percent in 2016. The Made in China 2025 plan aims to raise the domestic market share of Chinese-made robots to over 50 percent by 2020. At the same time, China's robot density was below the global average, with only 68 units per 10,000 workers in 2016, compared to the US stock of almost 200. Source: International Federation of Robotics (World Robotics Reports), Statista.com.



(a) Operational stock of industrial robots

(b) Labor share of automating firms

Figure 1: Automation and the labor share in China. Panel (a): operational stock of industrial robots in the five largest markets; China became the largest market in 2016. Source: International Federation of Robotics (2017), as reported in Cheng et al. (2019). Panel (b): value-added weighted labor share of automating firms in our sample, 2015 and 2017.

and can be applied to identify the impact of automation on the labor share whenever (exogenous) variation in prices is available.

Substantively, we find that the firm-level elasticity of substitution between automation capital and labor falls between 2.6 and 4.8 across different empirical specifications. Our preferred point estimate, which comes from the most credibly identified specification (based on city-industry average subsidies, net of the firm's own), is about 3.0. Because this elasticity exceeds one, automation capital and labor are strong substitutes among actively automating firms: cheaper automation lowers the firm's labor share. A unit elasticity, by contrast, would leave the labor share unchanged regardless of the price of automation. Our estimation decisively rejects that benchmark.

We then ask how much of the labor-share decline in our sample can be attributed to automation. Feeding the estimated elasticity into the calibrated model, we find that automation accounts for a substantial part of it. The aggregate effect is nonetheless smaller than the firm-level elasticity alone would suggest, because automating firms, though they cut their labor share sharply, account for only about a quarter of value added. The calibration also pins the elasticity from a second direction: a substantially lower value would imply counterfactually rapid growth in the value added of automating firms.

Related literature. Our paper is one of a few that estimate the causal impact of modern automation on the labor share. The most closely related studies are Acemoglu and Restrepo (2020) and Graetz and Michaels (2018). Both papers use different variants of a shift-share type of identification and do not focus on the labor share. Their estimates capture long-run general-equilibrium adjustments (price changes and factor reallocation) and so recover a blend of the technological elasticity and equilibrium feedback. We instead measure the firm-level technological elasticity directly, before any general-equilibrium adjustment muddies it. Because it is a structural primitive, our estimate can be fed into a model and carried to new counterfactuals; the two approaches are complementary. Unlike these studies, Autor and Salomons (2018) focus on the labor share itself, relating it to industry-level productivity growth, though without causally identifying automation's role.

A growing literature documents the firm-level consequences of robot adoption. Acemoglu et al. (2020) for France, Koch et al. (2021) for Spain, Bonfiglioli et al. (2024) for French robot imports, and Dixon et al. (2021) for Canada find that robot adoption raises firm output and productivity and reshapes the workforce, often lowering the adopter's labor share. We share this firm-level focus but differ in aim: rather than estimating reduced-form employment or labor-share responses, we use a policy-induced price instrument to recover the structural elasticity of substitution between automation capital and labor, a technological primitive that we then carry into general equilibrium. A complementary structural analysis is Humlum (2019), who studies robot adoption in Denmark and its distributional effects on employment.[6]

Our findings also speak to the broader debate over the declining labor share. The explanation closest to our mechanism is the falling price of capital emphasized by Karabarbounis and Neiman (2014): cheaper capital lowers the labor share when capital and labor are gross substitutes, and we provide direct firm-level evidence for substitution of exactly this kind, for the increasingly important case of automation capital. A second set of accounts moves the labor share through channels unrelated to automation technology: the rise of high-markup "superstar" firms (Autor et al., 2020), increasing market power and pure profits (De Loecker et al., 2020; Barkai, 2020), and the offshoring of labor-intensive production (Elsby et al., 2013). These are precisely the

[6]The work by Ciminelli et al. (2018) studies how employment protection deregulation has contributed to the decline in labor shares across advanced economies.

confounders our identification strategy is designed to purge. Finally, the elasticity we estimate is a different object from the capital-labor elasticity in this literature. We measure substitution between *automation* capital and labor, whereas Oberfield and Raval (2021) and others estimate substitution between capital broadly defined and labor, which lies below one both at the plant level and in the aggregate. Automation capital is precisely the form of capital that displaces labor, so a markedly higher elasticity is unsurprising.

The rest of the paper is organized as follows. Section 2 discusses the theory of automation that we use to develop our identification strategy. Section 3 discusses the data and our empirical strategy, and presents the results. Section 4 discusses the aggregate implications of our calibrated model and provides robustness analysis of our estimation procedure using model-generated data.

2 Model

We lay out the theory behind both our empirical and quantitative analysis. Our model is fairly standard and links the quality-adjusted price of automation capital to automation investment and the labor share.

2.1 Environment

Time is discrete and the horizon is infinite. There are n cities indexed by $c \in \{1, 2, \ldots, n\}$ and m industries indexed by $i \in \{1, 2, \ldots, m\}$. For convenience, we refer to the tuple (c, i) as an island. Islands are populated by a continuum of firms of a fixed measure $\Omega_{ci} > 0$, each producing a differentiated good that is aggregated into a composite homogenous consumption good sold at home and abroad at a unitary (global) price. Firms and goods are indexed by a unique identifier $\omega \in \Omega_{ci} \subset \mathbb{R}$, and we denote by $c(\omega)$ and $i(\omega)$ the city and industry of firm/good ω. There is an economy-wide labor market such that the wage rate is common to all firms. The economy is sufficiently small to take world prices as given and so the cost of funds is exogenous. To model shocks to firm-level markups (a major concern in the measurement of the causal link between automation and the labor share), we assume that a firm ω operates in one of k distinct markets, indexed by $j \in \{1, 2, \ldots, k\}$, on each island. The transition between these markets follows a Markov process independent of any firm or island characteristic, and the law of large numbers yields constant shares of firms in each market. We denote by $\Omega_{cij} \subset \Omega_{ci}$ the set of firms on island (c, i) that operate in market j. This structure allows us to demonstrate the robustness of our estimation procedure to the presence of such shocks.

Demand structure and aggregation

Goods are aggregated by two layers of competitive sectors: the final good sector and the intermediate good sector. On each island, the outputs of individual firms are combined into the production of an island-specific composite good and the island-specific composites are then again aggregated into a homogenous final good that is sold globally at a normalized (numeraire) price $P = 1$, with the goods market clearing condition dropped to reflect this assumption.

The final good producers convert a vector of differentiated island composite goods Q_{ci} into Y units of final goods according to the production function

$$Y = \left(\sum_{c=1}^{n} \sum_{i=1}^{m} D_{ci}^{\frac{1}{\rho}} Q_{ci}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}, \tag{1}$$

where D_{ci} are fixed weights such that $\sum_{ci} D_{ci} = 1$, and $\rho > 0$ is the elasticity of substitution between goods. Final good producers take prices as given and maximize static profits given by

$$\max_{Y, Q_{ci}} P(t) Y - \sum_{c=1}^{n} \sum_{i=1}^{m} P_{ci}(t) Q_{ci},$$

where P_{ci} is the price of the composite good from industry i in city c, and where Y is given by the production function (1). The resulting demand function for the composite good from island i is

$$Q_{ci}(t) = D_{ci} \left(\frac{P_{ci}(t)}{P(t)} \right)^{-\rho} Y,$$

and the normalization of the price of the final good implies

$$P(t) = \left(\sum_{c=1}^{n} \sum_{i=1}^{m} D_{ci} P_{ci}(t)^{1-\rho} \right)^{\frac{1}{1-\rho}} \equiv 1.$$

At the island level, a sector of competitive producers aggregates goods of all firms from all markets on the island into a composite bundle Q_{ci} sold to the final good sector at a unit price P_{ci}. The producers solve

$$\max_{q(\omega)} P_{ci}(t)\, Q_{ci} - \sum_{j=1}^{k} \int_{\Omega_{cij}} p(t,\omega)\, q(t,\omega)\, d\omega,$$

where $p(t,\omega)$ and $q(t,\omega)$ denote the price and quantity of good ω, respectively. That industry uses the production technology

$$Q_{ci} = \prod_{j=1}^{k} \left(\int_{\Omega_{cij}} d(t,\omega)^{\frac{1}{\theta_j}}\, q(t,\omega)^{\frac{\theta_j - 1}{\theta_j}}\, d\omega \right)^{\phi_j \frac{\theta_j}{\theta_j - 1}}.$$

The terms $d(t,\omega)$ are time-varying demand shocks affecting each individual firm and that follow an arbitrary process. The parameter $\theta_j > 1$ is the elasticity of substitution across goods in market j, and ϕ_j is the intensity of market j. We impose constant returns to scale on the production function and so $\sum_{j=1}^{k} \phi_j = 1$. By the zero profit condition, equilibrium prices satisfy

$$P_{ci}(t) = \prod_{j} \left(\frac{P_{cij}(t)}{\phi_j} \right)^{\phi_j},$$

and the price index for market j in city c in industry i at time t is

$$P_{cij}(t) = \left(\int_{\Omega_{cij}} d(t,\omega)\, p(t,\omega)^{1-\theta_j}\, d\omega \right)^{\frac{1}{1-\theta_j}}.$$

The demand curve for products produced by firm ω on island (c,i) and operating in market $j(t,\omega)$ is given by

$$q(t,\omega) = d(t,\omega) \left(\frac{p(t,\omega)}{P_{cij(t,\omega)}(t)} \right)^{-\theta_{j(t,\omega)}} \phi_{j(t,\omega)} \left(\frac{P_{ci}(t)}{P_{cij(t,\omega)}(t)} \right) Q_{ci}(t). \tag{2}$$

Notice that the market $j(t,\omega)$ the firm operates in, and which changes stochastically over time, shifts the overall demand for its goods (through ϕ_j) and the elasticity of that demand is θ_j.

Production technology

The production technology of a firm combines support capital k_s, equipment k_e, automation capital m, support labor l_s and production labor l, and it is summarized by the production function[7]

$$F_\omega\left(k_s, l_s, k_e, l, m\right) = A\left(t, \omega\right)\left(k_s^{\gamma_\omega} l_s^{1-\gamma_\omega}\right)^{\eta_\omega}\left(a_\omega^{\frac{1}{\sigma_\omega}}\left(k_e^{\alpha_\omega} l^{1-\alpha_\omega}\right)^{\frac{\sigma_\omega-1}{\sigma_\omega}} + \left(1-a_\omega\right)^{\frac{1}{\sigma_\omega}} m^{\frac{\sigma_\omega-1}{\sigma_\omega}}\right)^{\left(1-\eta_\omega\right)\frac{\sigma_\omega}{\sigma_\omega-1}}. \tag{3}$$

The block $k_s^{\gamma_\omega} l_s^{1-\gamma_\omega}$ corresponds to activities (tasks) in support of production activities, which are captured by the last term in parentheses. For instance, support capital k_s might include buildings and structures; support labor l_s could include management, sales force, or any support employees that assist production activities but whose employment is only indirectly affected by automation. These inputs are aggregated in a Cobb-Douglas fashion. The parameter $0 \leq \gamma_\omega \leq 1$ determines the intensity of capital and the parameter $0 \leq \eta_\omega \leq 1$ determines how intensive the firm is in support and productive activities. $A\left(t, \omega\right)$ is time-varying total factor productivity.

The second block of the production function (between the large parentheses) describes production activities. These activities can be completed by a mix of equipment k_e and production labor l, also combined in a Cobb-Douglas fashion, or by using labor-saving automation capital m. The first option captures traditional techniques for completing production activities and is characterized by a price-invariant labor share determined by the parameter $0 \leq \alpha_\omega \leq 1$. The second option captures automation and involves no labor. The elasticity of substitution $\sigma_\omega > 0$ links the two terms and determines how easy it is to automate activities that would traditionally be produced using an equipment-labor mix. The parameter $0 \leq a_\omega \leq 1$ describes how intensive the production function is in automation capital.

We use the parameter a_ω to distinguish between the intensive and extensive margins of automation investment. The *intensive* margin refers to the accumulation of automation capital m by firms that are able to use that equipment in production ($a_\omega < 1$). In contrast, the *extensive* margin of adoption (the spread of automation across firms) is determined by the distinction between firms that can use automation ($a_\omega < 1$) and those that cannot ($a_\omega = 1$).

The parameters of the production function in (3) are indexed by ω and they may vary across

[7] The production function pertains to the firm's value added and does not involve intermediate goods.

firms. We allow the distributions of these parameters to be arbitrary at this point, although the calibration of the model imposes more structure on how parameters are distributed.

To gain intuition about the notion of automation conveyed by the production function (3), consider a simple task that involves nailing two pieces of material together. This task can be done by a worker (production labor l) who uses a hammer (equipment k_e) or, alternatively, by an autonomous hammering robot (automation capital m) that performs the task on its own. The elasticity of substitution σ_ω then describes how easy it is for the firm to substitute production labor by automated machinery with any such "automatable" task. Finally, whoever does the hammering (worker or robot) might need a building (support capital k_s) as a place to work and a manager (support labor l_s) to supervise production, which is captured by the term in front of the bracket.[8]

A notable feature of the above production function is that it is consistent with the Kaldor (1961) fact that the labor share remained constant for several decades. Indeed, if automation capital is impossible to obtain (infinite price for m), (3) collapses to a standard Cobb-Douglas aggregator with a constant labor share equal to

$$LSO\left(t,\omega\right) := \left(\underbrace{\frac{\theta_{j(t,\omega)}-1}{\theta_{j(t,\omega)}}\left(1-\eta_\omega\right)\left(1-\alpha_\omega\right)}_{LSOP} + \underbrace{\frac{\theta_{j(t,\omega)}-1}{\theta_{j(t,\omega)}}\eta_\omega\left(1-\gamma_\omega\right)}_{LSON} \right). \tag{4}$$

But once the first automation technologies start to become available, firms might begin using them in production, in which case the labor share $LS\left(t,\omega\right)$ will move away from its initial $LSO\left(t,\omega\right)$ value. For later use, we refer to LSO as the automation-free labor share, $LSOP$ as the automation-free labor share in production activities and $LSON$ as the automation-free labor share in support (nonproduction) activities. By construction, automation only affects labor share via production activities pertinent to $LSOP$.

[8] Acemoglu and Restrepo (2020) microfound this interpretation of the above production function using a task-based model. In their theory, the parameter a_ω maps onto the number of tasks for which automation technology has been developed to date (automatable tasks), and the elasticity of substitution σ_ω maps onto the parameters pertaining to the task-level production function, which determines how much labor and capital the firm chooses per automatable task, given prices.

Firm problem

We denote the equilibrium wage rate by w. The user costs associated with the three forms of capital available to the firm are assumed to be determined by a fixed global cost of funds r, the price of each type of capital and their corresponding depreciation rates. We denote the user cost of structures by $r^s(t)$, the user cost of equipment by $r^e(t)$, and the user cost of automation by $r^m(t)(1-s(t,\omega))$, where $s(t,\omega)$ captures any policy-induced time-varying changes in the user cost of automation capital that may vary across firms. Note that $s(t,\omega)$ can vary across firms and across time. Later on, we will map $s(t,\omega)$ to government subsidies for automation investment that we observe in the data and use it to develop our identification strategy.

The user costs of capital are implied by our assumptions of fixed global cost of funds r, the assumption of linear depreciation of capital, and a fixed relative price of each type of capital good in terms of the global final good (which is the numeraire). For instance, if we denote by p^m the price of one unit of automation capital, then we can write

$$\underbrace{p^m(t)(1-s(t,\omega))(1+r)}_{\text{cost of funds for purchase}} - \underbrace{(1-\delta^m)p^m(t+1)(1-s(t,\omega))}_{\text{value of undepreciated capital}} = r^m(t)(1-s(t,\omega)), \tag{5}$$

where δ^m is the depreciation rate of automation capital and

$$r^m(t) := p^m(t)(1+r) - (1-\delta^m)p^m(t+1). \tag{6}$$

is the user cost of automation capital before subsidies. The formula assumes that the subsidy affects the cost of replacing undepreciated capital $(1-\delta^m)p^m$, which makes sense in the context of our data since the duration of "Made in China 2025" is about a decade and the estimated lifetime of automation equipment and robots is also about a decade. Under this assumption an increase in the subsidy rate s and a decline in the price of capital p^m are equivalent. User costs of other forms of capital are defined analogously to (6).

The problem of the firm boils down to a static profit maximization problem of the form

$$\pi(t,\omega) := \max_q (p(t,\omega) - \lambda(t,\omega))\, q, \tag{7}$$

where the price $p(t,\omega)$ satisfies the demand equation (2) and $\lambda(t,\omega)$ is the marginal cost of production, which is defined by the cost minimization problem of the form

$$\lambda(t,\omega) := \min_{l,l_s,k_e,k_s,m} r^s(t)\,k_s + r^e(t)\,k_e + r^m(t)\,(1-s(t,\omega))\,m + w(t)\,(l_s+l), \tag{8}$$

subject to $F_\omega(k_s,l_s,k_e,l,m) \geq 1$. A standard constant markup pricing rule applies as long as each firm is atomistic, and so

$$p(t,\omega) = \frac{\theta_{j(t,\omega)}}{\theta_{j(t,\omega)}-1}\lambda(t,\omega), \tag{9}$$

where $j(t,\omega)$ is the market in which firm ω currently operates. As expected, a change in market j is akin to a markup shock. The lemma below characterizes the marginal cost of production $\lambda(t,\omega)$ as a function of prices and parameters.

Lemma 1. *The marginal cost of production $\lambda(t,\omega)$ is given by*

$$\lambda(t,\omega) = \frac{1}{A(t,\omega)}\left(\frac{\lambda^s(t,\omega)}{\eta_\omega}\right)^{\eta_\omega}\left(\frac{\left(a_\omega\lambda^e(t,\omega)^{1-\sigma_\omega} + (1-a_\omega)\left(\lambda^m(t,\omega)\right)^{1-\sigma_\omega}\right)^{\frac{1}{1-\sigma_\omega}}}{1-\eta_\omega}\right)^{1-\eta_\omega}, \tag{10}$$

where

$$\lambda^s(t,\omega) = \left(\frac{r^s(t)}{\gamma_\omega}\right)^{\gamma_\omega}\left(\frac{w(t)}{1-\gamma_\omega}\right)^{1-\gamma_\omega},$$
$$\lambda^e(t,\omega) = \left(\frac{r^e(t)}{\alpha_\omega}\right)^{\alpha_\omega}\left(\frac{w(t)}{1-\alpha_\omega}\right)^{1-\alpha_\omega},$$
$$\lambda^m(t,\omega) = r^m(t)\,(1-s(t,\omega)).$$

Proof. All proofs are in Appendix D. □

The effect of the price of automation on the marginal cost is captured by the last term in between the large parentheses in (10) and it depends on the price of automation capital and the elasticity of substitution between automation capital and labor σ_ω.

Total labor supply in the economy is fixed and the wage rate w is determined by the usual economy-wide market clearing. The assumption of a global market for a homogeneous final good and common interest rate r eliminates the need for an explicit statement of the household problem and we omit it. The formal definition of equilibrium is standard and we also omit it.

2.2 Characterization of equilibrium labor share dynamics

We now provide a preliminary characterization of the impact of the price of automation capital on automation and the labor share. These equations are fundamental to the empirical strategy developed in the next section and provide the key intuition for the workings of the model.

We begin with a first result that relates the subsidy $s(t,\omega)$ to automation to labor ratio $m(t,\omega)/l(t,\omega)$.

Lemma 2. *The automation to labor ratio $m(t,\omega)/l(t,\omega)$ is given by*

$$\log \frac{m(t,\omega)}{l(t,\omega)} = \Theta(t,\omega) - \sigma_\omega \log(1 - s(t,\omega)), \tag{11}$$

where

$$\Theta(t,\omega) := \log \frac{1-a_\omega}{a_\omega} + \alpha_\omega \log \frac{\alpha_\omega}{1-\alpha_\omega} + \alpha_\omega \log \frac{w(t)}{r^e(t)} + \sigma_\omega \log \frac{\lambda^e(t,\omega)}{r^m(t)}. \tag{12}$$

Intuitively, the subsidy s_ω pushes the firm to acquire more automation capital m relative to the number of production labor l it employs. The magnitude of this effect depends on the elasticity of substitution σ_ω. Since the elasticity is constant, a 1 percent reduction in the price of automation capital, perhaps from a subsidy, is associated with a σ_ω percent reduction in automation capital per (production) worker, m/l.[9]

We next move to the labor share, which in the model is given by

$$LS(t,\omega) := \frac{w(t)(l(t,\omega) + l_s(t,\omega))}{y(t,\omega)}, \tag{13}$$

where $y(t,\omega) := p(t,\omega)q(t,\omega)$ denotes the output of firm ω (its value added).

[9] One advantage of normalizing automation capital m by production labor l (instead of, say, value added) is that the demand shock terms θ_j and ϕ_j do not appear in (11) and (12). The instrument would nonetheless be valid if such shocks were independent of $s(t,\omega)$ but it would not be possible to structurally identify parameter σ in that case.

The following lemma completes the characterization of the link between automation and the labor share.

Lemma 3. *The labor share $LS(t,\omega)$ of a firm depends on its automation to labor ratio $m(t,\omega)/l(t,\omega)$ through the expression*

$$\log \frac{LSO(t,\omega) - LS(t,\omega)}{LS(t,\omega) - LSN(t,\omega)} = \Psi(t,\omega) + \frac{\sigma_\omega - 1}{\sigma_\omega} \log \frac{m(t,\omega)}{l(t,\omega)}, \tag{14}$$

where $LSO(t,\omega)$ is given by equation (4),

$$LSN(t,\omega) := \frac{w(t)\, l_s(t,\omega)}{y(t,\omega)} = \frac{\theta_{j(t,\omega)} - 1}{\theta_{j(t,\omega)}} (1 - \gamma_\omega)\, \eta_\omega \tag{15}$$

is the labor share of support workers, which coincides with the automation-free support labor share $LSON$ from (4) because automation operates only within the production block, and where

$$\Psi(t,\omega) := \frac{1}{\sigma_\omega} \log \frac{1 - a_\omega}{a_\omega} - \alpha_\omega \frac{\sigma_\omega - 1}{\sigma_\omega} \log \left(\frac{\alpha_\omega}{1 - \alpha_\omega} \frac{w(t)}{r^e(t)} \right).$$

This lemma implies that automation capital per production-level employee (m/l), determined by the cost of automation by Lemma 2, affects the labor share. The left-hand side of (14) is a decreasing function of $LS(t,\omega)$, and so any increase in the right-hand side of the equation leads to a lower labor share. For instance, if automation capital m and the equipment-labor bundle $k_e^{\alpha_\omega} l^{1-\alpha_\omega}$ are substitutes ($\sigma_\omega > 1$), the adoption of automation technologies by the firm pushes the labor share down. As expected, the labor share is unaffected by investment in automation in the Cobb-Douglas case ($\sigma_\omega = 1$).

The terms $\Theta(t,\omega)$ and $\Psi(t,\omega)$ in the above lemmas show that the parameter a_ω has an analogous effect on the labor share to the subsidy s_ω, with the elasticity σ_ω determining the sensitivity of the labor share to changes in a_ω.

3 Empirical results

Having laid out our theory, we now turn to the data. We begin by describing the dataset and our sources. We then discuss how we use the two key equations derived in Lemmas 2 and 3 to identify the effect of automation on the labor share, and to structurally estimate the elasticity of substitution between labor and automation capital. Using government automation subsidies as an instrument, we find that the two are strong substitutes, with an elasticity of about three, so that automation significantly lowers a firm's labor share.

3.1 Data

Our data comes from the China Enterprise General Survey (CEGS).[10] The CEGS is a longitudinal large-scale study of manufacturing firms and workers in China conducted in three waves: 2015, 2016 and 2018. The 2018 wave covers five provinces across different geographic parts of China.[11] Because of its larger coverage, we use the 2018 wave, which retroactively provides data for the years 2015, 2016 and 2017 in a consistent format. Data collection has been meticulously done by a team of economists traveling to site.[12] There are 1,618 unique firms in our sample.[13]

In our analysis we use information on the wage bill and employment by type of workers and value added. One unique feature of the data is that it distinguishes between various types of capital equipment. We use data on investment in fully-automated industrial robots (Machine-1 in survey) and computer numerically controlled (CNC) semi-automated machinery (Machine-2), the information on the subsidies received from the government for the purchase of each type of machinery, and also the data on all other forms of capital (excluding structures), which we hereafter refer to as "ordinary capital" (Machine-3 in the survey).[14]

We define automation investment as the purchase of both Machine-1 and Machine-2 equip-

[10] The name of the survey changed in 2020 from the China Employer-Employee Survey (CEES).

[11] Firms were sampled from the third National Economic Census conducted in 2014. The sampling was conducted in two stages, each using probability proportionate-to-size sampling, with size defined as manufacturing employment. Therefore, the firm-level sample is representative in terms of employment size in China. Employees were surveyed with stratification: 6 to 15 employees were randomly selected in each firm, among which 2 to 3 were middle and senior managers.

[12] The survey design team informed us that for large firms it was not uncommon for the reviewers to stay on site for weeks to collect the data. In addition, hard data points, such as those pertaining to a firm's financials, have been taken from accounting records pulled on site.

[13] Overall, the response rate of firms was 2019/2417 = 83.5%. About 400 of these were dropped in data cleaning, leaving the 1,618 firms in our sample. We do not have the information about the exact procedure by which the raw data was cleaned by the data center in Wuhan. We were informed that the data center planned a public release of the cleaning procedure at a later date.

[14] The sample is designed to be representative of the Chinese manufacturing sector, but we do not have the weights needed to construct representative statistics at this point. As a result, our aggregated analysis pertains to the sample.

ment. A Machine-1 piece of equipment is an industrial robot as defined by ISO 8373: "an automatically controlled, reprogrammable multipurpose manipulator programmable in three or more axes, which may be either fixed in place or mobile for use in industrial automation applications." The Machine-2 category has been specifically designed for this survey to capture advanced labor-saving automation machinery that does not meet the stringent requirements of ISO 8373.

The key aspect of the survey that enables our analysis is that its timing overlaps with the first phase of "Made in China 2025," a vast government-led program that placed high-tech labor-saving automation technologies at the forefront of national industrial policy.[15] MIC introduced sizable subsidies paid to firms as a discount to the purchase price of automation capital. Our dataset provides detailed information on the payments of these subsidies between 2015 and 2017, including the type of equipment they target. Importantly, the implementation of MIC fell largely on local governments, which had some flexibility in the types of policies and subsidy rates to implement. As a result, we see a large amount of variation in subsidies across cities, industries and individual firms.[16] By October 2016, at least 70 provinces, cities and county-level administrations had released local MIC 2025 strategies with specific local priorities.[17] We exploit the variation

[15]While there are no official figures on the scale of the program, the goals are unprecedented and indicative of a sizable budget. For example, the objective of the program is to increase the adoption rate of automation from 33 percent of firms in 2015 to 64 percent in 2025. Other objectives are equally ambitious. Information about the program can be found at the official website of the State Council of China: http://www.gov.cn/zhuanti/2016/MadeinChina2025-plan/. The "Notice of the State Council on Made in China 2025" summarizes these goals as follows: "pilot the construction of smart factories/digital workshops in key areas, accelerate the application of technologies and equipment such as human-machine intelligent interaction, industrial robots, smart logistics management, and additive manufacturing in the production process, and promote the simulation and optimization of manufacturing processes, digital control, and status information real-time monitoring and adaptive control." (Translated from official document found at http://www.gov.cn/zhengce/content/2015-05/19/content_9784.htm using Google Translate.)

[16]One goal of this design was to foster experimentation to identify the most effective policies. In total, there were 30 pilot cities assigned by MIC 2025, fifteen of which are included in the CEGS Survey: Seven in Guangdong Province, five in Jiangsu Province, one in Hubei Province, one in Sichuan Province and one in Jilin Province. The experiments targeted ten industrial sectors: 1) next generation information technology; 2) robotics and advanced automatic machinery; 3) aerospace and aviation equipment; 4) maritime engineering equipment and advanced maritime vessel manufacturing; 5) advanced rail equipment; 6) new energy vehicles; 7) advanced electrical equipment; 8) agricultural machinery and equipment adoption; 9) new material; 10) biomedicine and high-performance medical devices.

[17]Appendix A provides three examples of MIC implementations, which show broad-based subsidy policy towards purchases of automation equipment. In 2015, MIC wasn't effectively in place because not many localities formulated their programs and MIC was introduced in the middle of the year. We do not have information about the exact timing of the introduction of the subsidies as for 2015, but we know that by the end of 2016 they were already introduced in most places.

in this policy to construct an instrument for automation investment. Specifically, as we explain below, we map subsidies onto $s(t, \omega)$ in our model and devise an IV strategy to identify the causal impact of automation on firm-level labor shares. Before we proceed, we provide an overview of the data.

3.2 Data structure and summary statistics

Tables 1 and 2 provide a preliminary characterization of the dataset. These tables group firms into those that report investment in automation in any year between 2015 and 2017 (automating firms) and those that report receiving subsidies for purchases of automation capital in any year during this time period (subsidized firms). We back out the subsidy rate by dividing the subsidy payments received from the government for the purchases of automation equipment by the total amount of automation purchases reported by firms, as opposed to using the actual policy variables for which we do not have complete information at the firm level. Hence, by definition, all subsidized firms in our data invest in automation capital.

Statistic	All firms	Automating firms	Subsidized firms
Number of cities	60	43	24
Number of industries	31	23	15
Number of city-industry pairs	652	119	39
Number of observations	4,449	490	136
Number of (unique) firms	1,558	170	47
Share in total employment (in %)	100%	24%	8.6%
Share in total value added (in %)	100%	25%	8.0%

Table 1: Sample Structure (all years 2015-2017)

As Table 1 shows, the data over the sample period 2015-2017 covers 60 cities and 31 industries, altogether 652 city-industry pairs and 1,558 firms (those of the 1,618 surveyed with the variables used in the analysis), amounting to 4,449 observations. Of those, there are 170 unique automating firms (with 490 observations in total) during this time period and they are found in 43 cities, 23 industries and 119 city-industry pairs. The sample of firms that report receiving subsidies for automation are spread out over 24 cities, 15 industries and 39 city-industry pairs, and amount to 47 firms (136 observations). In the majority of city-industry pairs in which there is at least one subsidized firm, the majority of automating firms report receiving subsidies (about 80 percent). While there are relatively few producers that automate, they are considerably larger and account for about a quarter of total value added and employment. Firms that report receiving subsidies for automation account for about a quarter of firms that automate.[18]

The fact that there are fewer subsidized firms than automating firms is a direct consequence of the MIC design. As the examples discussed in Appendix A illustrate, not all cities subsidize automation across all industries. The rules vary and they may prevent some firms from taking advantage of the subsidies in place. For instance, some cities require that industrial robots be domestically produced. Others have caps on the total budget of the program, so that some firms may be too late to receive a payment. We will later come back to the different reasons why firms within the same city-industry pairs might not receive the same subsidy when discussing our empirical strategy.

Table 2 provides summary statistics about the three groups of firms in our sample. As we can see from column 2 (N), all variables are well-populated, albeit not perfectly, and there is some variation in coverage across variables. Automating firms tend to be larger and subsidized firms are even larger. Both weighted and unweighted mean labor shares are declining over time, and their decline is more pronounced among subsidized firms.[19] The mean subsidy rate among subsidized firms is about 20 percent (22.8 percent weighted by value added, the rate used in the elasticity back-out below), and among automating firms it is below 7 percent. Subsidies and investment in automation are highly skewed, and most automating firms do not receive any

[18]Our definition of automating firms is narrower than the extensive margin of automation that may become active in the long run. Many firms that do not report any investment in automation between 2015 and 2017 do report having some stock of automation capital, and hence they too should be considered among firms that could automate. A firm may not be automating not only because it cannot automate but also because the time for automating is not good for idiosyncratic reasons (negative demand shock). Such a narrower definition of automating firms is nonetheless helpful because all relevant information from our study comes from these firms.

[19]Many firms appear to erroneously report their total labor share as the labor share of production workers, since at the same time they report that only a fraction of their employment are production workers, which would be inconsistent. Under the assumption of wages being equalized across production and nonproduction activities, our model implies that the labor share of production employees can be backed out from the share of employment in production. The mean value of this share in the data is .63, which also implies a labor share of production workers of .477*.63=.30, a value inconsistent with the raw data number of about .44. To rationalize this number would require higher wages among production workers than nonproduction workers, which we do not find plausible. To address this issue, we have dropped observations in which the labor share in production workers exceeds the total labor share, which gives $\overline{LSP} = 0.303$ and implies a labor share for non-production workers of $\overline{LSN} = .477 - .303 = .174$ in 2015. This figure is for all firms; the elasticity back-out below uses the corresponding value for automating firms in 2017, $\overline{LSN} = .148$.

subsidies for automation.[20]

Variable[1]		Centrality			Dispersion		Skewness
	N	W.Mean[2]	Mean	Median	S.D.	$P90-P10$	$\frac{P90-2P50+P10}{P90-P10}$
All firms							
Employment	4,686	956	310	101	478	895	.810
- % in production	4,686	.634	.628	.657	.181	.478	-.209
Auto. investment/VA[3]	709	.105	.175	.000	1.681	.253	1.00
LS '15 (labor share)	1,350	.477	.549	.540	.290	.799	-.0089
LS '17	1,516	.455	.533	.527	.281	.791	.011
LS '17−LS '15 (by firm)	1,317	−.024	−.019	−.007	.193	.361	-.052
LS of production employees '17	818	.303	.339	.311	.206	.545	.185
Automating firms							
Employment	511	1209	675	367	674	1814	.642
- % in production	511	.621	.624	.647	.156	.368	−.103
Auto. investment/VA[3]	472	.136	.264	.019	2.06	.365	.894
LS '15	157	.480	.543	.523	.290	.818	.055
LS '17	167	.456	.512	.468	.272	.764	.176
LS '17−LS '15	152	−.025	−.030	−.011	.172	.334	−.017
LS of production employees '17	101[4]	.308	.351	.326	.204	.527	.218
Subsidy rate	335	.085	.066	.000	.151	.219	1.00
Subsidized firms							
Employment	139	1370	890	824	712	1784	.157
- % in production	139	.639	.623	.659	.152	.362	-.282
Auto. investment/VA[3]	134	.309	.623	.077	3.794	.784	.803
LS '15	44	.467	.511	.483	.304	.814	.150
LS '17	47	.443	.437	.390	.253	.706	.222
LS '17−LS '15	44	−.037	−.080	−.024	.206	.254	−.456
LS of production employees '17	25	.323	.320	.325	.196	.577	.115
Subsidy rate	109	.228	.202	.117	.207	.519	.551

Table 2: Selected Summary Statistics

Notes: [1]Unless a specific year is noted, the values pertain to the average for the three-year period under consideration: 2015, 2016 and 2017. [2] Weighted mean is obtained by weighting each observation by a firm's share in total value added averaged across all years 2015-2017. [3] Value added is measured after subtracting all subsidies. [4] As described in the text, many firms appear to be instead reporting the total wage bill, and to deal with this issue we omit all observations when the two are equal to calculate this statistic. N means the number of observations. We first add nominal values for the three years without discounting and then take the ratios.

The decline in the labor share is correlated with investment in automation. Consider a simple

[20]We winsorize the top and bottom 5% of the sample variables. Whenever we construct a ratio we winsorize this ratio similarly. We discount by inflation such that nominal values are measured in 2015 RMB.

OLS specification of the form

$$\Delta LS_{\omega,t,t-1} = \alpha \log\left(\frac{x_{\omega,t-1}}{VA_{\omega,t-1}}\right) + \delta_\omega + \varepsilon_{\omega,t}, \tag{16}$$

where $\Delta LS_{\omega,t,t-1}$ is the change in labor share between two consecutive years, $t \in \{2015, 2016, 2017\}$, and $x_{\omega,t-1}$ denotes investment in machines by firm ω in year $t-1$, δ_ω is a firm fixed effect, and ε_t is an error term. To contrast the effect of automation and ordinary capital, we consider both automation investment (x_A) and other machine investment (x_O), the latter including all other capital excluding structures. The measures of investment in capital are normalized by value added (VA) to capture their importance compared to the size of the firm.[21]

It is clear from columns (1) and (2) in Table 3 that automation investment exhibits a negative and statistically significant association with the labor share. The size of the effect is also economically large. In contrast, columns (3) and (4) show that the relationship between investment in ordinary capital and the labor share cannot be significantly distinguished from zero.[22]

While these results show that the decline in labor share exhibits a higher partial correlation with investment in automation compared to that with investment in other machines, interpreting the automation investment coefficient in terms of structural parameters is fraught with problems. First, OLS does not address the issue of causality, i.e. changes in labor shares and investment in automation are endogenous responses of firms to changes in goods and factor prices, and so we need exogenous variation to provide evidence of a causal effect of automation on the labor share. Second, even with an instrument to uncover causality, we need a specification informed by a structural model in order to estimate key parameters of the model. The next section discusses how we address these challenges using subsidies for automation as an instrument in a 2SLS setup.

[21]We exclude from value added the subsidies paid by government for automation to avoid spurious correlation. Normalizing by production labor instead of value does not have a meaningful impact on the results.

[22]We test the robustness of the above relationship by using alternative specifications where we replace firm fixed effects with industry, city and year fixed effects but include commonly used time-varying firm specific attributes as control variables, namely, i) value added per worker $\left(\frac{VA}{l}\right)$, which controls for productivity; ii) capital per worker $\left(\frac{k}{l}\right)$, which controls for the capital intensity of a firm; iii) the ratio of total debt to total assets, which controls for the extent to which a firm is leveraged (LR); and iv) the export to total sales ratio $\left(\frac{X}{TS}\right)$, which controls for export intensity (higher export intensity tends to be associated with higher productivity, higher quality of products, more skilled workforce, and a greater product scope). All control variables are lagged by one period. The results are qualitatively the same and we report them in the appendix (Table 10). Export intensity, value added per worker, and leverage exhibit statistical significance.

Explanatory variables	Dependent Variable: $LS_{\omega,t} - LS_{\omega,t-1}$			
	Full Sample			
	(1)	(2)	(3)	(4)
$\log\left(\frac{x_A}{VA}\right)_{t-1}$	$-.0320$**	$-.0103$**		
	(.0160)	(.0044)		
$\log\left(\frac{x_O}{VA}\right)_{t-1}$			$-.0019$	.0006
			(.0179)	(0.0052)
Firm fixed effect	Yes	No	Yes	No
Year fixed effect	Yes	Yes	Yes	Yes
R-squared	0.81	0.03	0.58	.00
Observations	196	196	236	236

Table 3: Change in Labor Share and Automation Investment (OLS)

Notes: Least squares estimation. *** indicates significance at 1%, ** at 5% and * at 10% significance level. Standard errors in parentheses. Sample is restricted to observations with positive investment to ensure log values are well-defined. Value added calculation excludes government subsidies for investment that could bias the results.

3.3 Identifying the effect of automation: theory

The equations derived in lemmas 2 and 3 readily suggest a path to estimate the causal impact of automation on the labor share and the average elasticity $\mathbb{E}[\sigma_\omega]$ by using the observed variation in subsidy rates s_ω as an instrument.[23] Indeed, from (11) and (14) we know that s_ω affects the labor share only through its impact on the m/l ratio. Accordingly, a subsidy rate that is appropriately orthogonal to the firm's parameters and shocks can be used as a valid instrument to identify the effect of automation on the labor share. However, using these lemmas for the estimation is not possible in practice. First, we do not observe the terms $LSO(\omega, t)$ and $LSN(\omega, t)$ in the right-hand side of (14). As a result, we cannot compute the dependent variable in the second-stage regression. Second, we do not have reliable information on the *stock* of automation capital. We now show how this theory can be extended to estimate the effect of automation using the data that we have.

In what follows, let τ denote the three-year period 2015-2017 covered by the survey, and

[23]Throughout, we work with an expectation operator $\mathbb{E}[x]$ that is to be understood as the mean value of any variable or parameter x across all firms:

$$\bar{x} := \mathbb{E}[x] = \frac{1}{n}\sum_i \sum_c \int_{\omega \in \Omega_{ci}} x(t, \omega)\, d\omega,$$

and analogously for any conditional expectation operator.

$\tau - 1$ denotes the previous three-year period (which, in principle, we do not observe). Unless otherwise noted, we compute flows in period τ as the discounted sum of the flows for the years 2015, 2016 and 2017 (defined precisely in the next section). For stocks, we take the average of the three years. To calculate the change in the labor share across periods, lacking data for the preceding period, we take the difference between endpoints; that is, years 2017 and 2015. Such an approach, if anything, likely understates the change in the labor share and would work against finding a strong effect of automation.

Assumptions

To obtain identification, we impose two assumptions on the relationship between the subsidy rates and other elements of the model. We will discuss the empirical relevance of these assumptions in the next section.

Our first assumption is on the probabilistic structure of the subsidies.

Assumption 1. *The subsidy $s(\tau,\omega)$ follows the process*

$$s(\tau,\omega) = s_i(\tau) + s_c(\tau) + \varepsilon^s(\tau,\omega), \tag{17}$$

where s_i and s_c are industry- and city-specific mutually independent stochastic processes and $\varepsilon^s(\tau,\omega)$ is a mean-zero i.i.d. firm-specific stochastic process.[24]

Our second and key identifying assumption requires that the subsidy residual $\varepsilon^s(\tau,\omega)$ be orthogonal to other exogenous variables or parameters.

Assumption 2. *The random process $\varepsilon^s(\tau,\omega)$ is orthogonal to any parameter, shock or factor price (or their combination) $z(t,\omega)$ in the sense that*

$$\mathbb{E}\left[z(t,\omega)\,|\varepsilon^s(\tau,\omega)\right] = \mathbb{E}\left[z(t,\omega)\right]$$

for all t.

The above assumptions play a key role in our analysis. They ensure that the residual of the subsidy rate, after being projected onto city and industry fixed effects, satisfies the exclusion

[24]The proofs also assume finite moments as appropriate.

restriction for instrumental variable (IV) estimation. Note that the second assumption applies not only to the exogenous variables but also to the endogenous wage rate w, to industry-level price indices, as well as to the subcomponents of the subsidy: s_i and s_c in equation (17). This requires the subsidy program to be relatively small compared to the size of the relevant markets, including the labor market, so that the subsidy residual ε^s does not move these equilibrium objects "too much." Since the mean of the residual is zero, and only a fraction of firms are subsidized, we do not consider this to be a serious limitation. In the quantitative section we will nonetheless test the validity of this assumption using our calibrated model.

Furthermore, Assumption 2 implies that previous subsidy rates, which act as parameters, are similarly orthogonal to the residual $\varepsilon^s(\tau,\omega)$. While we view this assumption as reasonable, we show in Appendix E that the estimates presented below are conservative, in the sense that they provide a lower bound on the mean elasticity of substitution $\mathbb{E}[\sigma_\omega]$, so long as the residuals $\varepsilon^s(\tau-1,\omega)$ and $\varepsilon^s(\tau,\omega)$ are positively correlated.

Identification results

We next work out the link between a firm's investment in automation capital and the investment subsidy it faces. The result holds up to a first-order approximation (denoted by the symbol $\approx$) of the left-hand side of (14) and of the firm's policy functions with respect to the subsidy rate s. We also validate these approximations using numerical simulations of our model in the next section.

To be consistent with the institutional framework described in Section 3.2, we focus on investment decisions between two periods $\tau-1$ and τ, defined as

$$x(\tau,\omega) := p_m(\tau)\, m(\tau,\omega) - (1-\delta^m)\, p_m(\tau-1)\, m(\tau-1,\omega). \tag{18}$$

The following lemma relates investment $x(\tau,\omega)$ to the subsidy residual.

Lemma 4. *The automation investment intensity is related to the subsidy residual through the equation*

$$\mathbb{E}\left[\frac{x(\tau,\omega)}{l(\tau,\omega)}|\varepsilon^s(\tau,\omega)\right] \approx \mathcal{L}\varepsilon^s(\tau,\omega) + cte, \tag{19}$$

where $\mathcal{L}$ and cte are some constants.

This lemma is analogous to Lemma 2 and the same intuition applies, with the crucial difference that here it involves the investment rate x normalized by labor employed in production instead of the stock of automation capital m.

We now turn to our second result that links the conditional expectation of the change in a firm's labor share to the conditional expectation of its investment in automation.

Lemma 5. *The change in the labor share is related to automation investment intensity through the equation*

$$\mathbb{E}\left[LS\left(\tau,\omega\right)-LS\left(\tau-1,\omega\right)|\varepsilon^{s}\left(\tau,\omega\right)\right]\approx\mathcal{B}\mathbb{E}\left[\frac{x\left(\tau,\omega\right)}{l\left(\tau,\omega\right)}|\varepsilon^{s}\left(\tau,\omega\right)\right]+cte \tag{20}$$

where cte is a constant and where $\mathcal{B}$ is another constant given by

$$\mathcal{B}=-\frac{1}{1-\bar{s}}\frac{1}{\mathcal{L}}\left(\frac{1}{\overline{LS}-\overline{LSN}}+\frac{1}{\overline{LSO}-\overline{LS}}\right)^{-1}\mathbb{E}\left[\sigma_{\omega}-1\right], \tag{21}$$

where $\bar{s}$ is the mean subsidy rate across firms, $\overline{LS}$ is the average labor share, and $\overline{LSO}$ and $\overline{LSN}$ are the averages of (4) and (15), respectively.

The constant terms $\bar{s}$, $\overline{LS}$, $\overline{LSN}$ and $\overline{LSO}$ are the points around which first-order approximations are taken. We have some flexibility in choosing their values, but to minimize the approximation error they should be as close as possible to the actual observations, and the mean values are a natural choice.

The following proposition puts the two lemmas together to form the basis for our IV estimation in the next section.

Proposition 1. *The coefficients $\mathcal{B}$ and $\mathcal{L}$ can be consistently estimated from a two-stage least squares regression of the form:*

$$LS\left(\tau,\omega\right)-LS\left(\tau-1,\omega\right)\approx cte+\mathcal{B}\frac{x\left(\tau,\omega\right)}{l\left(\tau,\omega\right)}+FE_{i}+FE_{c}+e\left(\omega\right), \tag{22}$$

$$\frac{x\left(\tau,\omega\right)}{l\left(\tau,\omega\right)}\approx cte+\mathcal{L}s_{\omega}+FE_{i}+FE_{c}+u\left(\omega\right), \tag{23}$$

where e and u are error terms, $\mathcal{B}$ is given by (21) and FE_i, FE_c are a set of industry and city fixed effects, respectively.

This proposition shows that we can consistently estimate the causal impact of a change in automation investment on the labor share using the subsidy s_ω as an instrument. Furthermore, it implies that we can back out the average elasticity estimate from the estimated regression coefficients using the formula

$$\bar{\sigma} := \mathbb{E}\left[\sigma_\omega\right] = 1 - \mathcal{B}\mathcal{L}\left(1-\bar{s}\right)\left(\frac{1}{\overline{LS}-\overline{LSN}} + \frac{1}{\overline{LSO}-\overline{LS}}\right), \tag{24}$$

where the coefficients $\mathcal{B}$ and $\mathcal{L}$ come from estimating the system (22)–(23). The calculation requires values for $\bar{s}$, $\overline{LS}$, $\overline{LSN}$ and $\overline{LSO}$. We will describe in the next section how we come up with these numbers.

Measurement error

We briefly discuss the robustness of our strategy to measurement error in the investment intensity (x/l) and the subsidy rate.

Measurement error in the second-stage regressor x/l does not affect our estimate of $\mathbb{E}\left[\sigma_\omega\right]$, even when it is multiplicative rather than classical. Suppose we observe $\tilde{c} \times (x/l)$ for some orthogonal, positive-mean $\tilde{c}$ that satisfies Assumption 2. Because the elasticity is identified through the product $\mathcal{B}\mathcal{L}$, and $\tilde{c}$ scales the two stages in offsetting ways ($\mathcal{L}' = \mathbb{E}\left[\tilde{c}\right]\mathcal{L}$ and $\mathcal{B}' = \mathcal{B}/\mathbb{E}\left[\tilde{c}\right]$), the product $\mathcal{L}'\mathcal{B}' = \mathcal{L}\mathcal{B}$ is unchanged. Adding classical (additive) noise does not change this, since the IV removes it.[25]

Measurement error in the subsidy rate s is more consequential. If we observe $s' = \tilde{b} \times s$, the first-stage coefficient becomes $\mathcal{L}' = \mathcal{L}/\mathbb{E}\left[\tilde{b}\right]$, but the predicted values of x/l, and hence the second-stage coefficient $\mathcal{B}$, are unchanged, so the elasticity backed out from (24) is biased. Classical additive error in s is less of a concern: it attenuates the first stage, but the same attenuation reduces the variation in the predicted regressor and offsets in the second stage.

[25] If $\tilde{c}$ is a random variable that is correlated with x/l, however, we would only recover the coefficient in the second stage as long as the error term of the linear projection of $\tilde{c} \times (x/l)$ onto x/l has desirable properties, otherwise our estimation may be biased simply because the linearity assumption is not valid.

3.4 Estimation results

We use Proposition 1 to estimate the causal effect of automation investment by two-stage least squares. The dependent variable is the change $\Delta LS(\omega)$ in the labor share of a firm ω between 2015 and 2017.

Specification using firm-level subsidies

We begin with the most basic specification that uses the average subsidy rate calculated for each firm individually as an instrument for automation investment per production worker, x/l. We construct this subsidy rate by averaging the subsidy rates reported by each firm for the years 2015, 2016 and 2017. The subsidy rate for a firm in a given year is calculated by dividing the subsidy payments self-reported by the firm for purchases of automation capital by its total automation investment in that year.[26]

As for the variable we instrument for, we consider two specifications. The first adheres closely to the theory and uses total investment in automation over the entire sample period for each firm individually, deflated by the CPI and depreciation rate, and divides it by the corresponding average employment in production between the years 2015 and 2017. Formally, let δ_m be the depreciation rate of automation capital and let π_t be the CPI inflation in each year.[27] Let the cumulative investment in 2015 renminbi (RMB) between years $\underline{T}$ and $\overline{T}$ be

$$x_{\underline{T}-\overline{T}}(\omega) := \frac{\sum_{t=\underline{T},\ldots,\overline{T}} x(\omega,t)\left(\frac{1-\delta_m}{1+\pi_t}\right)^{\overline{T}-t}}{1+\overline{T}-\underline{T}},$$

where x_t is the surveyed firm's nominal spending on automation purchases in RMB. The first specification uses $x_{15-17}(\omega)/l_{15-17}(\omega)$ as instrument, where l_{15-17} is the average in production labor over those years.

Our second specification relies on the ratio $x_{17}(\omega)/l_{17}(\omega)$ instead, and effectively treats 2015

[26]Alternatively, one can take the sum of subsidies and investment for all years and then take the ratio. This specification leads to qualitatively similar coefficients. See the appendix (Table 11) for these results.

[27]We set $\delta_m = 10\%$ following Table C.1-5 "List of depreciation rates under the new asset code classification: Industrial machinery" from the Bureau of Economic Analysis. We use the Consumer Price Index (All Items for China, Index 2015=100, Annual, Not Seasonally Adjusted Values): CPI(2015) = 1, CPI(2016) = 1.02, CPI(2017) = 1.03625. Source: Economic Research Division, Federal Reserve Bank of St. Louis.

and 2016 as predetermined years. The reasoning behind this specification is that since MIC was announced in mid-2015 and effectively started to come online throughout 2016, investment rates prior to 2017 might not reflect the impact of the subsidies and, as a result, only add noise to the estimation of the first stage at the expense of not exactly adhering to the theory's implied relation in the second stage.[28] Depending on how important these errors are this may increase statistical power. As discussed, the IV deals quite well with measurement errors implied by not exactly adhering to the theory-implied law of motion.

The results of the 2SLS estimation for both specifications are shown in Table 4. We focus on the intensive margin of automation investment and therefore restrict the sample to firms with positive investment. In line with the model and Assumption (2), we include city and industry fixed effects, and so our identification relies on the variation in subsidies that are not captured by these fixed effects. We also include value added as a regressor to control for firm size.

In the first specification (columns 1 to 3), we find a negative and highly significant relationship between automation investment and the labor share. The F-statistics are also quite large, indicating that our instrument is strong. The relationship between automation investment and the labor share is also negative under the second specification (columns 4 to 6) although the relationship is somewhat weaker in this case.

The results of Table 4 are robust to various changes in the specification. In the appendix we show that the point estimates are stable when we drop the fixed effects (Table 12, which also shows that dropping firm size as a control has no meaningful impact) and when we add controls for state ownership and Party affiliation (Table 13). Finally, we check for heteroscedasticity in the error terms using the Pagan-Hall test. We find p-values close to 1 so that we cannot reject

[28] Investment decisions are planned ahead of time, and while some firms might have received a subsidy in 2016, it need not imply that the subsidy drove their investment decision back in 2016. Decisions in 2015 are unlikely to have been influenced at all by MIC since the program was announced in 2015. For this reason the first stage under the second specification may have more statistical power. The second stage will involve mismeasurement because it no longer adheres to the laws of motion implied by theory. For this measurement error not to become a problem it must be that the linear projection of one variable onto the other,

$$\frac{x_{17}}{l_{17}} = a_\omega \left(\frac{x_{16-17}}{l_{15-17}} \right) + a_i + a_c + e_\omega,$$

satisfies Assumption 2 (the error term e and the coefficient of the projection (e_ω, a_ω) must both be orthogonal to the subsidy residual). This follows from the second part of the proof of Proposition 1 (version of Frisch–Waugh–Lovell theorem), the discussion of measurement error in that section, and the fact that IV estimation removes the attenuation bias normally implied by the presence of classical measurement error e. We omit the formal proof.

the null hypothesis that the error terms are homoscedastic.

	(1)	(2)	(3)	(4)	(5)	(6)
	Second-stage dependent variable $\Delta LS_\omega = LS_\omega\,(2017) - LS_\omega\,(2015)$					
Automation investment (x_{15-17}/l_{15-17})	-0.0319^{***}	-0.0325^{***}	-0.0311^{***}			
	(0.0113)	(0.0113)	(0.0098)			
Automation investment (x_{17}/l_{17})				-0.0308^{*}	-0.0330^{*}	-0.0307^{**}
				(0.0158)	(0.0171)	(0.0156)
	First-stage dependent variable Automation investment (x/l)					
Subsidy rate (s)	13.48***	14.66***	14.84***	13.97***	14.43***	15.03***
	(3.271)	(2.584)	(3.130)	(4.061)	(4.265)	(4.924)
Industry / city fixed effects	No/Yes	Yes/No	Yes/Yes	No/Yes	Yes/No	Yes/Yes
Size (log VA)	Yes	Yes	Yes	Yes	Yes	Yes
# observations	148	148	148	148	148	148
F-statistic	16.99	32.19	22.48	11.83	11.44	9.319
$\bar{\sigma}$ ($\overline{LSO} = .60$)	4.38	4.75	4.63	4.38	4.75	4.63
	(0.831)	(0.855)	(0.806)	(0.831)	(0.855)	(0.806)
$\bar{\sigma}$ ($\overline{LSO} = .66$)	3.70	4.00	3.90	3.70	4.00	3.90
	(0.664)	(0.683)	(0.645)	(0.664)	(0.683)	(0.645)

Table 4: Impact of Automation on Labor Share using Firm-level Subsidies as Instrument

Notes: Two-stage least squares estimation (2SLS). *** indicates significance at 1%, ** at 5% and * at 10% significance level. Robust standard errors in parentheses. Sample is restricted to observations with positive automation investment. The instrument is the firm-level subsidy rate, the ratio of the subsidy received for purchases of automation capital (as defined in the text) to total investment over the sample period.

In all the specifications included in Table 4, the signs of the coefficients are consistent with an elasticity of substitution between labor and automation capital larger than one, so that higher automation investment has a negative and sizable effect on the labor share. A back-of-theenvelope calculation conveys the magnitude: using the average coefficients, a 1 percent subsidy (equivalently, a 1 percent decline in the price of automation equipment[29]) lowers an automating firm's labor share by about 0.46 percentage points (0.032×14.40), a large effect at the firm level.

This is a local, partial-equilibrium effect, and two caveats apply before reading anything aggregate into it. First, it omits the offsetting adjustment of prices and wages that dampens the response in equilibrium. Second, it pertains only to firms that actively invest in automation, which have seen larger labor-share declines than the average manufacturing firm. We quantify the aggregate impact properly with the calibrated model in the next section.

Structural interpretation of the estimated coefficients

We now combine the estimates from Table 4 with equation (21) to obtain a structural estimate of the average elasticity of substitution between labor and automation capital, given by the equation

$$\bar{\sigma} := \mathbb{E}\left[\sigma_\omega\right] = -\mathcal{B}\mathcal{L}\left(1-\bar{s}\right)\left(\frac{1}{\overline{LS}-\overline{LSN}}+\frac{1}{\overline{LSO}-\overline{LS}}\right)+1, \tag{25}$$

where, recall, $\mathcal{B}$ is the second-stage coefficient on automation investment, and $\mathcal{L}$ is the first-stage coefficient on the subsidy rate. Since our identification comes from automating firms, we approximate the expressions around $\bar{s} = 0.228$, which is the average subsidy across firms receiving a subsidy, and $\overline{LS} = 0.456$, which is the average labor share across automating firms in 2017. As for the labor share of nonproduction workers, we use $\overline{LSN} = 0.148$, which is the 2017 value for automating firms (see Table 2).[30]

Computing the average elasticity $\bar{\sigma}$ from (25) requires a value for $\overline{LSO}$, which, recall, would be the average labor share if the firms were not using any automation capital in production ($m = 0$). Given our estimated coefficients, the expression (25) is decreasing in $\overline{LSO}$, such that higher values of $\overline{LSO}$ are pushing for lower elasticities of substitution. Since $\overline{LSO}$ is a

[29] For quality-adjusted series of the price of industrial robots, see Graetz and Michaels (2018) (Figure 1). Their evidence points to a decline between 1990 and 2004 by a factor of five (based on 1990 dollars). For comparison, nominal wages grew on average 105 percent in the six reported countries. We are not aware of more recent quality-adjusted series but nonadjusted series are available from various industry sources.

[30] The standard errors we report for $\bar{\sigma}$ in Tables 4 and 5 are obtained by the delta method applied to the estimated coefficients $\mathcal{B}$ and $\mathcal{L}$, holding the back-out inputs $\bar{s}$, $\overline{LS}$, $\overline{LSN}$, and $\overline{LSO}$ fixed at the values stated here. They therefore reflect sampling uncertainty in the regression coefficients only and are conditional on these calibrated quantities.

counterfactual quantity, we cannot observe it directly. In the next section we adopt $\overline{LSO} = 60\%$ as our baseline and show that it is disciplined by the data: a substantially higher value would imply that automating firms were already heavily automated in 2015 and would overpredict the growth in their value added. Here, we also report estimates of $\mathbb{E}[\sigma_\omega]$ using the textbook value of the labor share of $\overline{LSO} = 66\%$. We view this number as quite conservative for two reasons. First, in the presence of profits, the value of $\overline{LSO}$ is bounded from above not by one but by a lower number that accommodates markups. Second, the manufacturing labor share in China was 57% in 1997, when very few, if any, modern industrial robots were at work.[31] On the other hand, the US labor share back in the 1960s was close to the textbook value. To the extent that US history is a better indicator, the value of $\overline{LSO} = 66\%$ is nonetheless appropriate.[32]

We report the estimates of $\mathbb{E}[\sigma]$ in the last rows of Table 4.[33] For the textbook value of the labor share, $\overline{LSO} = 66\%$, the average elasticity ranges from 3.7 to 4.0. For our baseline value of $\overline{LSO} = 60\%$ the analogous range is from 4.38 to 4.75. Importantly, all values are above 1, confirming that a decline in the price of automation equipment has a negative effect on the labor share. They are also relatively large, but in drawing such a conclusion one should keep in mind the magnitude of the decline in the US manufacturing labor share.[34] We return to the discussion of the estimated elasticity of substitution in the quantitative section.

Regression setup using city-industry average subsidies

The analysis so far has used each firm's own subsidy as the instrument. We now consider an alternative that relies only on city- and industry-level variation in the subsidy. We do so for two reasons. The first is exogeneity. Our firm-level instrument requires the firm-specific component of the subsidy, $\varepsilon^s(\tau, \omega)$, to be orthogonal to the firm's characteristics (Assumption 2). This could fail if subsidies are targeted at particular firms (whether by policymakers singling out firms for

[31]Data from the China Statistical Yearbook, 1997.

[32] Kehrig and Vincent (2021) report that the labor share in U.S. manufacturing declined from 61 percent in 1967 to 41 percent in 2012, which would suggest that historic evidence from the U.S. points to a higher value. By this measure, then, the textbook number of $\overline{LSO} = 66\%$ would be a conservative upper bound.

[33]Because $\bar{\sigma}$ depends on the regression estimates only through the product $\mathcal{BL}$ (which equals the reduced-form effect of the subsidy on ΔLS), specifications that share the same fixed effects and differ only in the automation measure (x_{15-17}/l_{15-17} versus x_{17}/l_{17}) deliver the same $\bar{\sigma}$ and the same standard error. The columns of Tables 4 and 5 that report identical elasticities reflect this invariance rather than a coincidence.

[34]Since the 1990s the US labor share index in manufacturing declined by 19% according to U.S. Bureau of Labor Statistics "Productivity and Costs" (1990-2018). See also discussion in footnote 32.

support or by firms lobbying on their own behalf), since the firm-specific subsidy would then be correlated with firm characteristics that also affect the labor share. An instrument built from the average subsidy of *other* firms in the same city and industry is immune to such firm-level targeting by construction.

The second reason is measurement. Investment decisions are affected by the subsidy rate that the firm *expects* to receive for these investments, not the subsidy that the firm actually received, and we do not observe these expectations. In the previous section we relied on the observed subsidy payments and assumed that what firms expected to receive equals what they actually received. But if firms which received zero subsidies ex post did not expect such an outcome ex ante, or firms which received full subsidy considered the risk of receiving zero subsidy ex ante, the measurement of the subsidy rate could induce a multiplicative bias that, as explained in the previous section, could adversely affect the estimated elasticity.[35]

To address both concerns, we consider an alternative instrument computed by averaging the subsidy rates reported by all the firms (except the one under consideration) in a given industry-city grouping, and then applying it to all firms for that grouping. This averages out the firm-specific component, removing any firm-level targeting, and, since the average realization of a random variable is closer to its ex ante expectation, also brings us closer to the expected policy subsidy. If the coefficients do not change, this is an indication that our estimates are immune to these problems. An additional advantage of this setup is that it increases the statistical power since more firms are "effectively" subsidized.

Formally, for any given city-industry pair (c, i) and year $t \in \{2015, 2016, 2017\}$, we construct the instrument for a firm ω as follows:

$$s_{\omega}^{\Omega_{ci}} = \frac{\sum_{t=2015}^{2017} \sum_{\omega \in \Omega'_{ci}} S_{ci}(t, \omega)}{\sum_{t=2015}^{2017} \sum_{\omega \in \Omega'_{ci}} x_{ci}(t, \omega)},$$

where $S_{ci}(t, \omega)$ is the subsidy transfer (in RMB) deflated by the CPI inflation and reported by firm ω from city c and industry i in period t, $x_{ci}(t, \omega)$ is automation investment similarly deflated

[35]It is possible that firms are simply misreporting the subsidy payments that they receive for their investments. It is also possible that, when making investment decisions, firms were facing some uncertainty about the size of the subsidy payments that they would receive. In either case, averaging the firm-level rates across all firms in a city-industry pair might get us closer to the true policy subsidy that was expected by the firms.

by CPI; the set $\Omega'_{ci} \subseteq \Omega_{ci}$ includes all automating firms from city-industry pair (c, i). To back out the average elasticity in this case, we use the same $\bar{s} = 0.228$ and otherwise the analysis is unchanged.

The regression results are presented in Table 5 and are overall similar to those reported in Table 4. The standard errors are, however, somewhat wider, so the elasticity estimates are correspondingly less precise. In our preferred specification that uses city and industry fixed effects (column 6), we still find a negative and highly significant coefficient in the second-stage regression, so that automation investment is still associated with a lower labor share. Compared to Table 4, the estimated elasticities $\overline{\sigma}$ are noticeably smaller under these specifications (around three rather than four and a half), though still comfortably above one. We adopt this column as our preferred estimate, which yields $\overline{\sigma} \approx 3.0$ for $\overline{LSO} = .60$ (and 2.6 for the textbook value $\overline{LSO} = .66$). We favor the city-industry average subsidy, net of the firm's own, over the firm-level subsidy of Table 4 because it is less susceptible to firm-level endogeneity in the realized subsidy, at the cost of wider standard errors on the elasticity.[36]

For completeness, we ran an analogous exercise for ordinary capital (Machine-3), for which we also have capital-type specific subsidy information. We found the coefficients to be insignificant. Together with our previous OLS results, this indicates an important difference between ordinary and automation capital: while we find that automation capital has a clear negative impact on the labor share, there is no evidence that ordinary capital has a similar effect.[37]

3.5 Limitations and robustness

We conclude this section by discussing further limitations of our instrument and discuss how we addressed them.

First, it is possible that subsidies during the sample period may be correlated with some other subsidies that were previously in place. Our theory, recall, assumes that preexisting subsidies are uncorrelated with the residual of the current set of subsidies after controlling for city and industry fixed effects. While we are unaware of any previous major national initiative that specifically

[36]Controlling for the size of the firms using value added does not have a meaningful impact on the results.

[37]The lack of significance in this case may indicate that the instrument is not relevant and it should not necessarily be interpreted as evidence against any relationship between price of capital and the labor share.

	(1)	(2)	(3)	(4)	(5)	(6)
	Second-stage dependent variable $\Delta LS_\omega = LS_\omega\,(2017) - LS_\omega\,(2015)$					
Automation investment (x_{15-17}/l_{15-17})	−0.0249*	−0.0317**	−0.0214**			
	(0.0138)	(0.0149)	(0.0091)			
Automation investment (x_{17}/l_{17})				−0.0148*	−0.0222*	−0.0122***
				(0.0076)	(0.0115)	(0.0047)
	First-stage dependent variable Automation investment (x/l)					
Subsidy rate ($s_\omega^{\Omega_{ci}}$)	10.50***	8.490***	12.10***	17.63***	12.10***	21.26***
	(3.224)	(2.711)	(3.949)	(3.689)	(3.677)	(4.531)
Industry / city fixed effects	No/Yes	Yes/No	Yes/Yes	No/Yes	Yes/No	Yes/Yes
Size (log VA)	Yes	Yes	Yes	Yes	Yes	Yes
# observations	136	136	136	136	136	136
F-statistic	10.61	9.809	9.385	22.83	10.83	22.03
$\bar{\sigma}$ ($\overline{LSO} = .60$)	3.06	3.11	3.04	3.06	3.11	3.04
	(1.060)	(0.930)	(1.082)	(1.060)	(0.930)	(1.082)
$\bar{\sigma}$ ($\overline{LSO} = .66$)	2.64	2.69	2.63	2.64	2.69	2.63
	(0.847)	(0.743)	(0.866)	(0.847)	(0.743)	(0.866)

Table 5: Impact of Automation on Labor Share using City-industry Average (minus firm i) Subsidies as Instrument

Notes: Two-stage least squares estimation (2SLS). *** indicates significance at 1%, ** at 5% and * at 10% significance level. Robust standard errors in parentheses. Sample is restricted to observations with positive automation investment. The instrument is the leave-out city-industry average subsidy rate: the mean firm-level subsidy rate in firm i's city-industry grouping, excluding firm i itself.

targeted automation investment on that scale, other economic policies were in place in China before MIC 2025 and it is possible that some small subsidies for automation were also in place. We investigate how this would affect our results in Appendix E and find that, in the likely case of a positive correlation between the residuals of the pre-existing subsidies and subsidies under MIC

2025, our estimation of the average elasticity $\bar{\sigma}$ would be biased downward, providing a lower bound for the true elasticity. Since we find high values to begin with, this does not change the conclusions from our findings. Intuitively, a positive correlation reduces the link between current subsidies and labor share changes as the firm would have already responded to the subsidies.

Second, subsidy programs may have targeted specific firms or industries whose labor shares were changing for reasons unrelated to their investment in automation, which would lead our instrument to identify a causal link where none exists. As discussed above, our preferred leave-out instrument is by construction immune to the targeting of *individual* firms; the residual concern is targeting at the industry or city-industry level. We have read through official documentation and found no evidence of subsidies being used to target firms or industries in this way. The possibility remains because subsidized firms tend to be larger, but controlling for size does not change the regression results, and we show in the appendix (Table 13) that controlling for party connections and government ownership does not invalidate them either.

4 Quantitative results

Section 3 estimated a firm-level elasticity of substitution between automation capital and labor of about three. That elasticity governs how a single firm's labor share responds to a change in the price of automation. It does not, by itself, tell us how the *aggregate* labor share moves, because aggregate movements also reflect the reallocation of value added across firms that automate to differing degrees. Nor does it tell us whether the reduced-form strategy of Section 3, which rests on a linear approximation and on asymptotic arguments, recovers that elasticity reliably in a sample as small and as heterogeneous as ours.

We take up both questions with the calibrated model. We find, first, that automation accounts for a substantial part of the decline in the labor share between 2015 and 2017, though less than the firm-level elasticity alone would suggest, because automating firms produce only about a quarter of value added. We find, second, that our identification strategy is sound: the instrument recovers the true elasticity in model-generated data, and the linear approximation that makes the estimator operational introduces only a modest bias, small enough that the central conclusion, that automation capital and labor are strong substitutes, is unaffected.

Throughout this section, unless noted otherwise, means across firms are weighted by value added.

4.1 Calibration

The model is the one laid out in Section 2. We simulate an economy that mirrors the city-industry structure of the survey: 44 cities, 23 industries, and the 122 city-industry groupings that contain at least one automating firm.[38] We solve the model on a discretized version of this structure: each grouping holds a large number of firms, to mimic the continuum. Only when we later validate the estimator on model-generated data (Section 4.3) do we enlarge the economy and draw subsamples the size of the survey. Our interest is in automating firms, but we also include firms that cannot automate ($a_\omega = 1$), so that the model can speak to the whole sample; these firms matter only through the general-equilibrium feedback between automation and prices.

We assume that within each grouping firms with the highest demand are the ones that automate. A fully specified model would derive adoption from a profit-maximizing extensive-margin decision, which we do not undertake; our rule is a reasonable reduced form, since larger firms find automation worthwhile, and it generates exactly the dependence between automation and firm demand that the subsidy instrument is designed to purge. Appendix F collects the details.

The model of Section 2 lets every technology parameter differ across firms; the calibration restricts this heterogeneity, both for parsimony and because the survey moments cannot identify more. We assume that η_ω, the importance of support relative to production activities, varies across firms, following a beta distribution.[39] We hold the two capital intensities equal and common, $\alpha = \gamma$: under a uniform markup the available moments cannot separately identify their dispersion from that of η, and equating them also makes the mapping from the labor share to the technology parameters transparent. Firms differ in two further respects already present

[38] The calibration is built from a cell-level summary of the survey (counts and averages by city-industry cell), prepared separately from the firm-level regression panel of Section 3. Minor differences in coverage make its counts slightly larger (122 groupings and 177 automating firms, against 119 and 170 in the regression sample of Table 1), and the two are not meant to coincide.

[39] We normalize its support to $[.05, .95]$ to keep η away from the extremes.

in the model: their capability to automate ($a_\omega < 1$ versus $a_\omega = 1$) and a demand shifter $d(\omega, \cdot)$ that generates the dispersion in firm size and its growth. Productivity A is common to all firms; in particular, automating firms receive no productivity premium, so they are large because they draw high demand, not because they are more productive. This is a deliberate choice, with consequences for the model's fit that we return to in Section 4.2.

A few parameters are then set outside the estimation. We fix the markup at a conventional 33 percent for every firm, which implies a common θ solving $\theta/(\theta-1) = 1.33$, or $\theta = 4.03$.[40] We adopt the elasticity of substitution from the preferred specification of Section 3, the city-industry 2SLS with both fixed effects (last column of Table 5), and take it to be common across firms, sectors and cities. Through (24) this elasticity depends on the automation-free labor share $\overline{LSO}$: we adopt $\overline{LSO} = .60$, which gives $\sigma = 3.04$, and carry the textbook alternative $\overline{LSO} = .66$, which gives $\sigma = 2.63$, as a robustness case throughout.

The same choice of $\overline{LSO}$ pins the technology intensities and the mean of η. The value added-weighted means of the technology parameters (which for the homogeneous α and γ are simply their common values) satisfy, from the labor-share identity (4),

$$\alpha = \gamma = 1 - \frac{\theta}{\theta - 1}\overline{LSO}, \qquad \overline{\eta} = 1 - \frac{\overline{LSOP}}{\overline{LSO}}.$$

Using the labor shares of automating firms (Table 2), this gives $\alpha = \gamma = .20$ and $\overline{\eta} = .25$ at $\overline{LSO} = .60$. We compute user costs of capital from (5) under standard assumptions collected in the appendix.

A simulated method of moments chooses the remaining parameters (the dispersion of η, the dispersion of the demand shifter in each period, the level and change of the automation price p_m, and the trend growth of productivity A) to match six moments jointly. Table 6 lists the moments and the resulting parameter values. Every moment responds to every parameter, but each is most informative about one: the dispersion of the production-employment share and of the labor-share components pin the spreads of η and of demand; the initial labor share of automating firms pins the initial automation price, and hence the automation capital these

[40]The survey does report a profit share (operating income over value added, about 34 percent), but that figure is gross of the cost of capital, whereas the model's profit share $1/\theta$ is net of it; the two are different objects, and the estimated elasticity is in any case insensitive to the markup, so we use the standard value.

firms had already accumulated by 2015; the dispersion of value-added growth pins the demand innovation; the near-doubling of the aggregate automation stock between 2015 and 2017 pins the fall in the automation price; and manufacturing wage growth pins productivity growth.[41]

				Value	
Targeted moment	Data	Model	Parameter	$\overline{LSO}$=.60	$\overline{LSO}$=.66
c.v. $l(\omega,\tau)/L(\omega,\tau)$	.25	.27	$\zeta_1(\eta), \zeta_2(\eta)$	.53, 1.90	.47, 1.98
c.v. $LSN(\omega,\tau)/LSP(\omega,\tau)$	1.88	1.80	$\sigma_{d(\tau-1)}$	1.08	1.11
w.m. $LS(\omega,\tau-1)$, autom.	.48	.48	$p_m(\tau-1)$	6.26	5.71
s.d. $\Delta_{\%} y(\omega,\tau)$	.53	.53	$\sigma_{d(\tau)}$	.42	.42
$\Delta_{\%} \sum_\omega m(\omega,\tau)$	98%	98%	$p_m(\tau)/p_m(\tau-1)$	.979	.962
$w_\tau/w_{\tau-1}$	12.5%	12.5%	$A_\tau/A_{\tau-1}$	1.065	1.068
markup	33%	33%	θ	4.03	4.03
elasticity (Sec. 3)			σ	3.04	2.63

Table 6: Calibration: targeted moments and parameter values

Notes: The top block lists the six moments matched jointly by the simulated method of moments and, for each, the parameter it most influences at the calibrated values. The bottom block lists the two parameters set outside the estimation: the markup and the elasticity adopted from Section 3; the latter, through (4), also pins the means of the technology parameters reported in the text. Both calibrations appear: the baseline $\overline{LSO} = .60$ and the robustness case $\overline{LSO} = .66$. "w.m." denotes the value added-weighted mean, "s.d." the standard deviation, "c.v." the coefficient of variation (the latter two unweighted), and $\Delta_{\%}$ the percentage growth rate. Appendix F defines the moments and details the calibration.

The baseline $\overline{LSO} = .60$ lies below the textbook labor share of .66, and the difference is disciplined by the data. A higher $\overline{LSO}$ would, through (24), imply a lower elasticity; it would also imply that automating firms were already heavily automated in 2015, since their automation-free labor share would have had to fall further to reach the observed level of .48. Combined with the targeted doubling of the automation stock, a higher $\overline{LSO}$ therefore implies a larger expansion in the value added of automating firms than we see in the data. The robustness column reports the model under the higher value; the appendix discusses this mechanism in more detail.

4.2 The aggregate effect of automation

The reduced-form analysis identified the firm-level elasticity σ. That elasticity predicts how automation moves a firm's own labor share, but the aggregate labor share also depends on how value added shifts between firms that automate to different degrees. We now use the calibrated

[41] The automation stock follows industrial-robot density in Chinese manufacturing, which rose 98 percent between 2015 and 2017 (International Federation of Robotics). Wage growth of 12.5 percent is real manufacturing wage growth over the same period (National Bureau of Statistics of China, deflated by the all-items CPI).

model, which captures both margins, to ask whether the estimated micro elasticity is consistent with the aggregate changes in the sample.

Table 7 compares the calibrated model to the data for automating firms, subsidized firms, and all firms. The model reproduces the labor share of automating and subsidized firms in 2015 and tracks its decline through 2017. The labor share of automating firms falls from .48 to .45 in the model, a drop of .037 against .025 in the data; the model somewhat overstates the decline because the largest firms adjust their labor share more sluggishly than it predicts. Subsidized firms, a subset of automating firms, are the heaviest automators, so both their automation investment and their labor-share decline exceed the automating-firm average; the model reproduces both patterns.

Statistic		Firm type					
		Automating		Subsidized		All	
		Data	Model	Data	Model	Data	Model
Labor share in $\tau - 1$	$LS\left(\omega, \tau-1\right)$ w.m.	.480	.482	.467	.481	.477	.566
Labor share in τ	$LS\left(\omega, \tau\right)$ w.m.	.456	.445	.443	.417	.455	.556
Change in labor share	ΔLS_ω w.m.	-.025	-.037	-.037	-.065	-.024	-.011
Automation investment[1]	$\sum_\omega p_m x_\omega / \sum_\omega y_\omega$	15.2%	15.2%	22.6%	23.9%	3.6%	4.8%

Table 7: The aggregate effect of automation in the calibrated model

Notes: The baseline calibration ($\overline{LSO}$ = .60, σ = 3.04) against the data, by firm type. "w.m." denotes the value added-weighted mean. [1]The absolute scale of automation investment is not pinned by the moments (it moves one-for-one with the normalization of the automation price), so we report it normalized to the automating-firm investment rate of 15.2 percent observed in the data.

The central lesson of Table 7 is that automation accounts for a substantial part of the labor-share decline, though less than the firm-level elasticity alone would suggest. In the calibrated model the value added-weighted labor share of all firms falls by .011, nearly half of the .024 decline in the data. Two forces hold the aggregate response below the firm-level elasticity. First, automating firms produce only about a quarter of value added, so even the sizable .037 fall in their own labor share moves the aggregate by less. Second, the model assigns the entire decline to automating firms, whereas in the data non-automating firms also see their labor share fall; gauged directly, the automating firms' own decline accounts for about a quarter of the aggregate drop ($.25 \times .025 \approx .006$ against the observed .024). Either way, automation is a quantitatively important driver of the falling labor share, without implying an excessive aggregate sensitivity. The lower-elasticity robustness calibration ($\overline{LSO} = .66$, $\sigma = 2.63$) tells the same story, with a slightly larger labor-share decline for automating firms ($-.043$; subsidized $-.070$; all firms $-.013$).

The model is built to capture automating firms, and it reaches its limits on three dimensions, all detailed in the appendix. First, it overstates their value-added share, .31 against .25 in the data. Second, it understates the dispersion of their size. Both follow from making automating firms large through demand alone, with no productivity premium and a uniform markup. Third, it overstates the aggregate labor share, .57 against .50, because the non-automating firms that produce most of value added are held at the automation-free benchmark $\overline{LSO} = .60$ rather than at their lower measured level. The model thus matches the behavior of automating firms, not the level of the aggregate labor share.

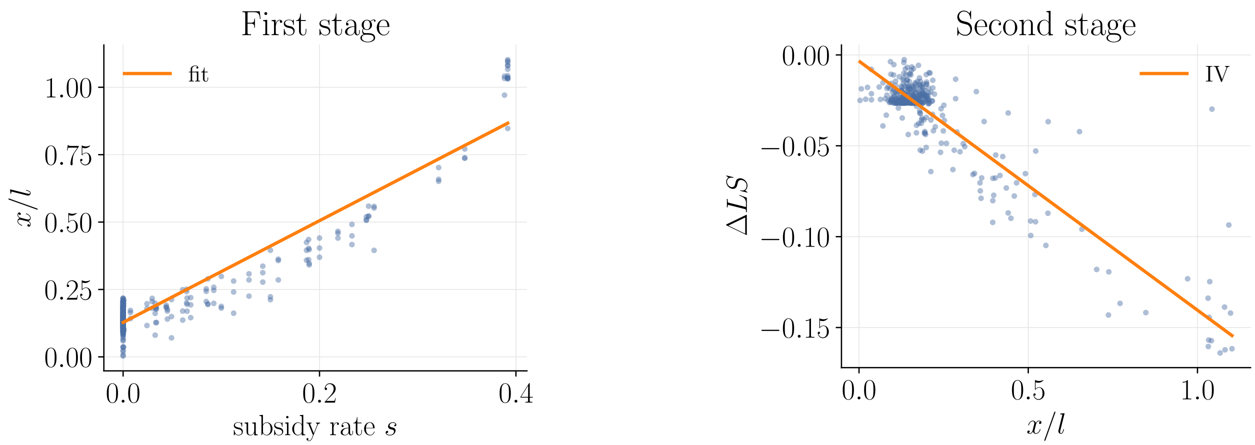
4.3 Validation of the empirical strategy

The estimator of Section 3 rests on Proposition 1, which is valid under two idealizations: a linear approximation of the firm's policy functions, and an asymptotic argument applied to a sample of only 177 automating firms. The relationship the estimator approximates is in fact curved, and 177 firms is not many. We use the calibrated model, where the true elasticity is known, to check how much these two departures matter.

The exercise mirrors the empirical one. We solve the model in general equilibrium on a large economy, with many firms per grouping, so that its population moments are effectively the truth; we then run the Section 3 estimator on subsamples that match the survey's 177 automating firms and their distribution across groupings. The large economy isolates the bias of the estimator, and the subsampling adds back the noise of a small one.

Figure 2 shows the estimator at work on data from the calibrated economy. The first stage (panel a) relates a firm's automation-to-labor ratio to its subsidy and is strongly upward-sloping, so the instrument has power. The second stage (panel b) relates the change in the labor share to the instrumented ratio; its slope is the instrumental-variable estimate from which we recover the elasticity. We can then compare that estimate to the value we put into the model.

Table 8 reports the elasticity the estimator recovers, in the large-economy population and in 177-firm subsamples, against the true value of 3.04. It returns 2.80 in the population and 2.77 in the small sample, an under-recovery of about eight to nine percent. The shortfall in the



(a) First stage: x/l against the subsidy

(b) Second stage: ΔLS against the instrumented x/l

Figure 2: The estimator at work on data generated by the calibrated model: a strong first stage (panel a), and the second-stage relationship whose slope identifies σ (panel b).

population is from the curvature that the linear approximation neglects; the further slippage in the sample is the cost of working with 177 firms. Both are small.

	Population	Sample ($n = 177$)
True (calibrated) σ	3.04	3.04
Recovered elasticity (Section 3 estimator)	2.80	2.77 (0.16)

Table 8: Elasticity recovered from model-generated data

Notes: The elasticity recovered by the Section 3 estimator from data generated by the calibrated model, whose true elasticity is 3.04. "Population" applies the estimator to the large-economy general equilibrium; "Sample" averages it over subsamples of 177 automating firms matching the survey's distribution across groupings, the figure in parentheses being the sampling standard error across resampled subsamples. Recovered values average over 16 simulated economies.

We read this wedge as modest rather than correcting for it. Its size, and even its sign, depend on the cross-firm dispersion of labor shares, which the uniform-markup model understates, so the precise figure is not a structural constant to invert. What matters is the magnitude: the estimator recovers the elasticity to within about ten percent, and the estimate is reasonably precise, with a sampling standard error of about 0.16 in a single 177-firm sample, so the recovered elasticity sits many standard errors above one. Accounting for the approximation and the small sample therefore changes the economics little: across every cut the recovered elasticity remains far above one, and the conclusion that automation capital and labor are strong substitutes stands.

Appendix G reports the full validation, including a homogeneous benchmark that isolates the role of the instrument and an exact, data-infeasible estimator confirming that identification is sound.

5 Conclusion

We have developed a new methodology to estimate the impact of automation on the labor share and have used it on micro data from China. We found a negative and large causal impact of automation on the labor share of automating firms. We also estimated an elasticity of substitution between labor and automation capital of about 3.0, the key structural parameter that describes how the labor share is affected by the price of automation. Our findings suggest that further declines in the price of industrial robots may lead to a sizable redistribution of a firm's income away from workers and toward capital owners among automating firms. The same force appears to have been at work beyond China. We do not study the United States directly, but the strong substitutability we estimate offers a natural channel for the long decline in its manufacturing labor share, from about .61 to .41 between 1967 and 2012, over decades of steadily falling automation prices.[42] One has to be cautious in translating our results to aggregate effects. Throughout the paper, we have focused on the intensive margin of automation investment, i.e. how much firms that are already automating react to changes in the price of robots. Future research could also explore the role of the extensive margin to better understand how automation technologies are initially introduced in a workplace. Another important question we leave unanswered is how our findings extend to other sectors of the economy, outside manufacturing.

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[42] The U.S. manufacturing labor-share figures are from Kehrig and Vincent (2021); while that analysis does not explicitly link the decline to automation, it rules out several other explanations. On the secular decline in the quality-adjusted price of automation equipment, see Graetz and Michaels (2018).

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Appendices

A The Made in China 2025 program

The announcement of MIC 2025 was followed by the release of Made in China 2025 Major Technical Roadmap (2015), also known as Green Book, which specifies explicit targets on growth rate and market share, and details various policy assistances for each targeted industry and its sub-sectors. Though there are four categories to organize policy action and targets - innovation, quality and efficiency, digitization of industry, and environment protection -, we report the two most relevant from the standpoint of automation.[43]

Category	Index	2013	2015	2020	2025
Quality and efficiency	Manufacturing quality competitive index	83.1	83.5	84.5	85.5
	Growth of manufacturing value-added			2% higher than 2015 average	4% higher than 2015 average
	Growth of manufacturing productivity			7.5% growth in 2015-2020	6.5% growth in 2020-2025
Digitization of industry	Broadband coverage index (%)	37	50	70	82
	Information and digitization technology adoption index (%)	52	58	72	84
	Automation usage index (%)	27	33	50	64

Notes: 1) manufacturing quality competitive index is based on 12 specific indexes. 2) broadband coverage index = the number of families using broadband / total number of families. 3) Information technology adoption index = the number of firms adopting information and digitization technology / total number of firms (calculation based on a 30,000 random above-scale enterprises sample). 4) Automation usage index refers to the average automation adoption rate among all above-scale enterprises

Table 9: MIC 2025 Goals in Manufacturing Sectors

Selected examples of implementation

Here we provide three examples of how automation subsidies under MIC have been implemented by different cities. The information is based on official program descriptions.

Foshan (Guangdong Province): "assign 130 million RMB per year to support automation and robotic machinery [...] the government provides subsidies on robots purchases: 12% of machines' value if the robots are made in Foshan (the maximum subsidy cannot be larger

[43] More detail can be found at http://www.gov.cn/zhengce/content/2015-05/19/content_9784.htm.

than 3 million RMB per year); 8% of machines' value if the robots are made elsewhere (the maximum subsidy cannot be larger than 2 million per year) [...] every year, the government awards 8 million RMB per firm to 10 selected firms as the automation demonstration based on the following criteria [...]"[44,45]

Huzhou (Zhejiang Province): "to encourage automation, the government provides subsidies based on the following three categories: a) 6% of machines' value (one time claim cannot be larger than 10 million RMB) for four industries: metal material, furniture, modern textile, and fashion products; b) 8% of machines' value (one time claim cannot be larger than 15 million RMB) for three industries: information technology, advanced machinery, and biomedicine. c) 10% of machines' value (one time claim cannot be larger than 20 million RMB) for the industrial areas of integrated circuit, new energy (including adoption to electronic vehicles, battery, and machinery), logistics equipment, aerospace and aviation equipment, new medical technology.[46]

Wuhan (Hubei Province): "the government provides subsidies at the rate 12% of machines' value (one time claim cannot be larger than 3 million)."

As we can see from these examples, some cities offer blanket subsidies for automation while others target specific sectors of activity.[47]

B Variable definitions

Value-added (VA): Sum of four components: 1) employees' wages and compensations; 2) depreciation of fixed assets; 3) net taxes on production; 4) business earning surplus. This approach follows the standard value-added calculation from the National Bureau of Statistics of China (unit: 10,000 RMB) but excludes subsidies that enter our analysis. Importantly, government subsidies are not part of value-added to avoid polluting the construction of the labor share variable. **Robots investment** (I_R): Purchase value of robots (unit: 10,000 RMB). **Automatic machine investment** (I_{AM}): The purchase value of CNC and other automatic and

[44]These criteria are "a) the firm's business registration is in the city of Foshan; b) the plant is in the city of Foshan; c) the annual sales of main business is at least 50 million RMB; d) total investment on automation is at least 30 million RMB; e) at least 60% of machines' investment is for automation; f) the firm development strategy needs to be consistent with national industrial policy."

[45]Text from the Bureau of Industry and Information Technology of Foshan City available at http://fsiit.foshan.gov.cn/.

[46]Text from the Bureau of Economy and Information of Huzhou City available at http://hzjx.huzhou.gov.cn/.

[47]Text from the official government website of Wuhan City at http://www.wuhan.gov.cn/.

semi-automatic machines (unit: 10,000 RMB). **Other machine investment** (I_{OM}): The purchase value of other machines (unit: 10,000 RMB). **Robots subsidy** (S_R): The amount of subsidy for purchasing robots (unit: 10,000 RMB) **Automatic machine subsidy** (S_{AM}): The amount of subsidy for purchasing CNC and other automatic and semi-automatic machines (unit: 10,000 RMB). **Other machine subsidy** (S_{OM}): The amount of subsidy for purchasing other machines (unit: 10,000 RMB). Employment (L): **Total employees** (unit: person). **Lerner index** (LI): Operating income (without taxes) to value added. **Labor share** (ls): The ratio of labor wages and compensation to value-added. **Labor share of production employees** (also production employment share) (lsp) The ratio of labor wages and compensation of production workers to value-added. **Export intensity** (X/VA): The ratio of export sales to value-added.

C Robustness checks

C.1 OLS: Change in Labor Share and Automation Investment

OLS Robustness to Inclusion of Firm-Level Characteristics

The table below shows the OLS results for the case where we drop firm fixed effects but include industry and city fixed effects and firm-level characteristics.

C.2 Additional IV Estimation Results

IV Estimation with Alternate Measure of Firm Level Subsidy Rate

The instrument is the subsidy rate at the firm-level calculated by taking the ratio of total subsidy received for purchases of automation capital (as defined in text) over the three years to the total investment over the three years.

IV Estimation without Fixed Effects

Firm ownership and party affiliation

In this appendix, we augment the main specification of Table 4 to control for party affiliation (dummy variable that equals 1 if the owner is a party member and 0 otherwise) and ownership status of the firm (dummy variable that equals 1 if the firm is state owned and 0 otherwise).

Dependent Variable: $\Delta LS_{i,t,t-1}$		
Explanatory variables	(1)	(2)
$\log\left(\frac{x_A}{VA}\right)_{t-1}$	-0.331**	
	(0.164)	
$\log\left(\frac{x_O}{VA}\right)_{t-1}$		-0.107***
		(0.022)
$\log(LR)_{t-1}$	-0.023	-0.087**
	(0.015)	(0.042)
$\log\left(\frac{X}{TS}\right)_{t-1}$	-0.016*	-0.048**
	(0.009)	(0.022)
$\log\left(\frac{VA}{l}\right)_{t-1}$	-0.086***	-0.11
	(0.021)	(0.063)
$\log\left(\frac{K}{l}\right)_{t-1}$	0.021	-0.028
	(0.014)	(0.04)
Year fixed effect	Yes	Yes
Industry fixed effect	Yes	Yes
City fixed effect	Yes	Yes
Firm fixed effect	No	No
R-squared	0.442	0.715
Observation	960	1079

Table 10: Change in Labor Share and Automation Investment - Firm-Level Characteristics

Notes: Least squares estimation. *** indicates significance at 1%, ** at 5% and * at 10% significance level. Standard errors in parentheses. Sample is restricted to observations with positive investment to ensure log values are well-defined.

	Second stage dependent variable $\Delta LS_\omega = LS_\omega\,(2017) - LS_\omega\,(2015)$
Automation investment (x/l)	-0.0315***
	(0.010)
	First stage dependent variable Automation investment (x/l)
Subsidy rate (s)	14.46***
	(3.284)
Industry / city fixed effect	Yes/Yes
Size (log value added)	Yes
# observations	149
R^2 (2nd stage)	0.375
F-statistic	19.39

Table 11: Impact of automation on labor shares using firm-level subsidies as an instrument - Alternate Subsidy Rate.

Notes: Two-stage least squares estimation (2SLS). *** indicates significance at 1%, ** at 5% and * at 10% significance level. Robust standard errors in parentheses. Sample is restricted to observations with positive automation investment. The instrument is the subsidy rate at the firm-level calculated by taking the ratio of total subsidy received for purchases of automation capital (as defined in text) over the three years to the total investment over the three years.

	(1)	(2)
	Second stage dependent variable $\Delta LS_\omega = LS_\omega\,(2017) - LS_\omega\,(2015)$	
Automation investment (x/l)	-0.0289**	-0.0296**
	(0.0138)	(0.0133)
	First stage dependent variable Automation investment (x/l)	
Subsidy rate (s)	11.64***	12.16***
	(2.816)	(2.894)
Industry / city fixed effect	No/No	No/No
Size (log value added)	No	Yes
# observations	148	148
F-statistic	17.09	17.65

Table 12: Impact of automation on labor shares using firm-level subsidies as an instrument – No Fixed Effects.

Notes: Two-stage least squares estimation (2SLS). *** indicates significance at 1%, ** at 5% and * at 10% significance level. Robust standard errors in parentheses. Sample is restricted to observations with positive automation investment. The instrument is the subsidy rate at firm-level calculated by taking the ratio of subsidy received for purchases of automation capital (as defined in text) and total investment for each year and then averaging over the three years of the sample period.

	(1)	(2)
	Second stage dependent variable $\Delta LS_\omega = LS_\omega\,(2017) - LS_\omega\,(2015)$	
Automation investment (x/l)	-0.0318***	-0.0313***
	(0.00976)	(0.0095)
	First stage dependent variable Automation investment (x/l)	
Subsidy rate (s)	15.16***	15.07***
	(3.017)	(3.20)
Industry / city fixed effect	Yes/Yes	Yes/Yes
Party affiliation	Yes	No
Government ownership	No	Yes
Size (log value added)	Yes	Yes
# observations	147	146
R^2 (2nd stage)	0.378	0.375
F-statistic	25.26	22.14

Table 13: Impact of automation on labor shares using firm-level subsidies as an instrument and including ownership and party affiliation controls.

Notes: Two-stage least squares estimation (2SLS). *** indicates significance at 1%, ** at 5% and * at 10% significance level. Robust standard errors in parentheses. Sample is restricted to observations with positive automation investment. The instrument is the firm-level subsidy rate.

D Omitted proofs and derivations

This appendix includes derivations and additional results related to the model of Section 2.

Proof of Lemma 1:

The first-order conditions corresponding to the cost minimization problem (8) are

$$\begin{aligned}
k_s &: r^s(t)\,k_s = \lambda\gamma_\omega\eta_\omega \\
l_s &: wl_s = \lambda(1-\gamma_\omega)\,\eta_\omega \\
k_e &: r^e(t)\,k_e = \lambda a_\omega^{\frac{1}{\sigma_\omega}}\alpha_\omega(1-\eta_\omega)\left(k_e^{\alpha_\omega}l^{1-\alpha_\omega}\right)^{\frac{\sigma_\omega-1}{\sigma_\omega}}\Gamma^{-1} \\
m &: r^m(t)\,(1-s_\omega)\,m = \lambda(1-a_\omega)^{\frac{1}{\sigma_\omega}}(1-\eta_\omega)\,m^{\frac{\sigma_\omega-1}{\sigma_\omega}}\Gamma^{-1} \\
l &: wl = \lambda a_\omega^{\frac{1}{\sigma_\omega}}(1-\alpha_\omega)(1-\eta_\omega)\left(k_e^{\alpha_\omega}l^{1-\alpha_\omega}\right)^{\frac{\sigma_\omega-1}{\sigma_\omega}}\Gamma^{-1}
\end{aligned} \tag{26}$$

where we define

$$\Gamma_\omega \equiv a_\omega^{\frac{1}{\sigma_\omega}}\left(k_e^{\alpha_\omega}l^{1-\alpha_\omega}\right)^{\frac{\sigma_\omega-1}{\sigma_\omega}} + (1-a_\omega)^{\frac{1}{\sigma_\omega}}\,m^{\frac{\sigma_\omega-1}{\sigma_\omega}}. \tag{27}$$

and where λ is the Lagrange multiplier on the constraint $F_\omega(k_s, l_s, k_e, l, m) \geq 1$. Because of the constant returns to scale, the Lagrange multiplier λ equals the firm's marginal cost. Combining the first-order conditions, it is straightforward to show that (10) holds and we omit an explicit derivation. Q.E.D.

Proof of Lemma 2:

We use the first-order conditions derived in the proof of Lemma 1. Combining the first-order conditions with respect to k_e and l, we have

$$\left(k_e^{\alpha_\omega}l^{1-\alpha_\omega}\right)^{\frac{1}{\sigma_\omega}} = \lambda a_\omega^{\frac{1}{\sigma_\omega}}(1-\eta_\omega)\left(\frac{\alpha_\omega}{r^e}\right)^{\alpha_\omega}\left(\frac{1-\alpha_\omega}{w}\right)^{1-\alpha_\omega}\Gamma_\omega^{-1}, \tag{28}$$

which, together with the first-order condition with respect to m, gives

$$\left(\frac{m}{k_e^{\alpha_\omega}l^{1-\alpha_\omega}}\right)^{\frac{1}{\sigma_\omega}} = \left(\frac{1-a_\omega}{a_\omega}\right)^{\frac{1}{\sigma_\omega}}\left(\left(\frac{\alpha_\omega}{r^e}\right)^{\alpha_\omega}\left(\frac{1-\alpha_\omega}{w}\right)^{1-\alpha_\omega}\right)^{-1}\left(r^m(1-s_\omega)\right)^{-1}.$$

From the first-order conditions with respect to k_e and l we also have $\alpha_\omega wl = (1-\alpha_\omega) r^e k_e$, and so the previous expression becomes

$$\frac{m}{l} = \left(\frac{1-a_\omega}{a_\omega}\right)\left(\frac{\alpha_\omega}{1-\alpha_\omega}\frac{w}{r^e}\right)^{\alpha_\omega}\left(\left(\frac{\alpha_\omega}{r^e}\right)^{\alpha_\omega}\left(\frac{1-\alpha_\omega}{w}\right)^{1-\alpha_\omega}\right)^{-\sigma_\omega}\left(r^m\left(1-s_\omega\right)\right)^{-\sigma_\omega},$$

which corresponds to (11) after simplifications. Q.E.D.

Proof of Lemma 3:

The labor share of support (nonproduction) workers is

$$LSN\left(t,\omega\right) := \frac{w\left(t\right) l_s\left(t,\omega\right)}{y\left(t,\omega\right)} = \frac{w\left(t\right) l_s\left(t,\omega\right)}{\frac{\theta_{j(t,\omega)}}{\theta_{j(t,\omega)}-1}\lambda\left(t,\omega\right) q\left(t,\omega\right)} = \frac{\theta_{j(t,\omega)}-1}{\theta_{j(t,\omega)}}\left(1-\gamma_\omega\right)\eta_\omega,$$

where the last equality follows from the first-order condition for labor (scaled up to an arbitrary output q) from Lemma 1. Similarly for production labor we can compute the labor share $LSP\left(t,\omega\right) := \frac{w(t)(t,\omega)}{y(t,\omega)}$ so that $LS\left(t,\omega\right) = LSN\left(t,\omega\right) + LSP\left(t,\omega\right)$ by the definition of the labor share (13).

By combining the (scaled) first-order condition with respect to l with the definition of Γ (see Lemma 1) and the fact that $\alpha_\omega wl = (1-\alpha_\omega) r^e k$ and $p = \frac{\theta_j}{\theta_j - 1}\lambda$, we have that

$$\begin{aligned} LSP\left(t,\omega\right) := \frac{w\left(t\right) l\left(t,\omega\right)}{y\left(t,\omega\right)} = {} & \frac{\theta_{j(t,\omega)}-1}{\theta_{j(t,\omega)}}\left(1-\eta_\omega\right)\left(1-\alpha_\omega\right) \\ & \times\left(1+\left(\frac{1-a_\omega}{a_\omega}\right)^{\frac{1}{\sigma_\omega}}\left(\frac{1-\alpha_\omega}{\alpha_\omega}\frac{r^e}{w}\right)^{\alpha_\omega\frac{\sigma_\omega-1}{\sigma_\omega}}\left(\frac{m\left(t,\omega\right)}{l\left(t,\omega\right)}\right)^{\frac{\sigma_\omega-1}{\sigma_\omega}}\right)^{-1} \end{aligned}$$

or, equivalently,

$$\frac{1}{LSP\left(t,\omega\right)} = \left(\frac{\theta_{j(t,\omega)}-1}{\theta_{j(t,\omega)}}\left(1-\eta_\omega\right)\left(1-\alpha_\omega\right)\right)^{-1}\left(1+\left(\frac{1-a_\omega}{a_\omega}\right)^{\frac{1}{\sigma_\omega}}\left(\frac{1-\alpha_\omega}{\alpha_\omega}\frac{r^e}{w}\right)^{\alpha_\omega\frac{\sigma_\omega-1}{\sigma_\omega}}\left(\frac{m\left(t,\omega\right)}{l\left(t,\omega\right)}\right)^{\frac{\sigma_\omega-1}{\sigma_\omega}}\right).$$

Combining with $LS\left(t,\omega\right) = LSN\left(t,\omega\right) + LSP\left(t,\omega\right)$ and using the definition of $LSO\left(t,\omega\right)$ in equation (4) yields the result. Q.E.D.

Proof of Lemma 4:

From (18), we can write

$$\frac{x(\tau,\omega)}{l(\tau,\omega)} = p_m(\tau)\frac{m(\tau,\omega)}{l(\tau,\omega)} - (1-\delta^m)\, p_m(\tau-1)\frac{m(\tau-1,\omega)}{l(\tau-1,\omega)}\frac{l(\tau-1,\omega)}{l(\tau,\omega)}.$$

Using (11), the last expression becomes

$$\frac{x(\tau,\omega)}{l(\tau,\omega)} = p_m(\tau)\exp(\Theta(\tau,\omega))(1-s_\omega)^{-\sigma_\omega} \tag{29}$$

$$- p_m(\tau-1)(1-\delta^m)\exp(\Theta(\tau-1,\omega))(1-s_{\omega,-1})^{-\sigma_\omega}\frac{l(\tau-1,\omega)}{l(\tau,\omega)}, \tag{30}$$

where $\Theta(\tau,\omega)$ is given by (12) and where $s_{\omega,-1} := s(\tau-1,\omega)$.

We handle the two terms on the right-hand side of that equation separately. Taking a first-order approximation of the first term with respect to s_ω around some level $\bar{s}$ we find

$$\exp(\Theta(\tau,\omega))(1-s_\omega)^{-\sigma_\omega} \approx \exp(\Theta(\tau,\omega))(1-\bar{s})^{-\sigma_\omega} + \sigma_\omega \exp(\Theta(\tau,\omega))(1-\bar{s})^{-\sigma_\omega-1}(s_\omega-\bar{s}).$$

As for the second term, we linearize it with respect to s_ω to obtain

$$p_m(\tau-1)(1-\delta^m)\exp(\Theta(\tau-1,\omega))(1-s_{\omega,-1})^{-\sigma_\omega}\frac{l(\tau-1,\omega)}{l(\tau,\omega)} \approx \mathcal{N}(\tau-1,\tau,\omega) + \mathcal{H}(\tau-1,\tau,\omega)(s_\omega-\bar{s}),$$

where $\mathcal{N}$ and $\mathcal{H}$ are terms that depend on the linearization point but that are not functions of s_ω itself (just $\bar{s}$); that is $\mathcal{N}$ and $\mathcal{H}$ only depend on exogenous variables such as parameters and current level of shocks (other than s_ω). Here note that the linearization error is introduced only by linearization with respect to s_ω and not the value of the shocks; in other words, we do not linearize with respect to shock values and only take a linear approximation around their current value with respect to s_ω.

Putting these expressions back into (29) we find

$$\frac{x(\tau,\omega)}{l(\tau,\omega)} = p_m(\tau)\exp(\Theta(\tau,\omega))(1-\bar{s})^{-\sigma_\omega}$$
$$+ \big(p_m(\tau)\sigma_\omega\exp(\Theta(\tau,\omega))(1-\bar{s})^{-\sigma_\omega-1} - \mathcal{H}(\tau-1,\tau,\omega)\big)(s_\omega-\bar{s}) - \mathcal{N}(\tau-1,\tau,\omega).$$

Plugging in $s_\omega = s_i + s_c + \varepsilon^s_\omega$ from (17), taking expectation of both sides conditional on ε^s_ω, and using the assumed probability structure of subsidies together with Assumption 2, gives

$$\mathbb{E}\left[\frac{x\left(\tau,\omega\right)}{l\left(\tau,\omega\right)}|\varepsilon_{\omega}^{s}\right]=cte+\mathbb{E}\left[p_{m}\left(\tau\right)\sigma_{\omega}\exp\left(\Theta\left(\tau-1,\tau,\omega\right)\right)\left(1-\bar{s}\right)^{-\sigma_{\omega}-1}-\mathcal{H}\left(\tau-1,\tau,\omega\right)\right]\varepsilon_{\omega}^{s},$$

where *cte* is a constant. Q.E.D.

Proof of Lemma 5:

We simplify the notation by writing $\varepsilon_{\omega}^{s}:=\varepsilon^{s}\left(\tau,\omega\right)$, which is without loss under Assumption 2. A first-order approximation of (11) around a point $\bar{s}$ yields

$$\log\left(\frac{m\left(\tau,\omega\right)}{l\left(\tau,\omega\right)}\right)\approx c_{i}\left(\tau,\omega\right)+\frac{\sigma_{\omega}}{1-\bar{s}}s\left(\tau,\omega\right), \tag{31}$$

where $c\left(\tau,\omega\right):=\Theta\left(\tau,\omega\right)-\sigma_{\omega}\log\left(1-\bar{s}\right)+\frac{\sigma_{\omega}}{1-\bar{s}}\bar{s}$ includes all the constants. Combining (14) with (31) yields

$$\log\frac{LSO\left(\tau,\omega\right)-LS\left(\tau,\omega\right)}{LS\left(\tau,\omega\right)-LSN\left(\tau,\omega\right)}=\Psi\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{\sigma_{\omega}}c_{i}\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{1-\bar{s}}s\left(\tau,\omega\right). \tag{32}$$

Now computing the difference of that equation between the periods τ and $\tau-1$ and taking the conditional expectation, we get

$$\mathbb{E}\left[\log\frac{LSO_{\tau}-LS_{\tau}}{LS_{\tau}-LSN_{\tau}}-\log\frac{LSO_{\tau-1}-LS_{\tau-1}}{LS_{\tau-1}-LSN_{\tau-1}}|\varepsilon_{\omega}^{s}\right]=cte+\mathbb{E}\left[\frac{\sigma_{\omega}-1}{1-\bar{s}}\right]\varepsilon_{\omega}^{s}, \tag{33}$$

where we have again used Assumption 2 and the probability structure (17) of the subsidy policy, and where we have abbreviated $LSO\left(\omega,\tau\right)\equiv LSO_{\tau}$ and so on. Now, note that the left-hand side of that expression can be written as

$$\begin{aligned}\log\frac{LSO_{\tau}-LS_{\tau}}{LS_{\tau}-LSN_{\tau}}-\log\frac{LSO_{\tau-1}-LS_{\tau-1}}{LS_{\tau-1}-LSN_{\tau-1}}&=\log\left(LS_{\tau-1}-LSN_{\tau-1}\right)\\&-\log\left(LS_{\tau}-LSN_{\tau}\right)\\&+\log\left(LSO_{\tau}-LS_{\tau}\right)\\&-\log\left(LSO_{\tau-1}-LS_{\tau-1}\right),\end{aligned} \tag{34}$$

and that we can approximate each term using the Taylor expansion

$$\log\left(x-y\right)\approx\log\left(\bar{x}-\bar{y}\right)+\frac{1}{\bar{x}-\bar{y}}\left(x-\bar{x}\right)-\frac{1}{\bar{x}-\bar{y}}\left(y-\bar{y}\right).$$

So, for instance, we obtain

$$\log\left(LS_{\tau-1}-LSN_{\tau-1}\right)\approx\log\left(\overline{LS}-\overline{LSN}\right)+\frac{1}{\overline{LS}-\overline{LSN}}\left(LS_{\tau-1}-\overline{LS}\right)$$
$$-\frac{1}{\overline{LS}-\overline{LSN}}\left(LSN_{\tau-1}-\overline{LSN}\right).$$

Repeating this argument on all the terms on the right-hand side of (34), we find that it can be written as

$$\begin{aligned}\log\frac{LSO_{\tau}-LS_{\tau}}{LS_{\tau}-LSN_{\tau}}-\log\frac{LSO_{\tau-1}-LS_{\tau-1}}{LS_{\tau-1}-LSN_{\tau-1}}&\\ =\left(\frac{1}{\overline{LS}-\overline{LSN}}+\frac{1}{\overline{LSO}-\overline{LS}}\right)\left[LS_{\tau-1}-LS_{\tau}\right]&\\ -\left(\frac{1}{\overline{LSO}-\overline{LS}}\right)\left[LSO_{\tau}-LSO_{\tau-1}\right]&\\ +\frac{1}{\overline{LS}-\overline{LSN}}\left[LSN_{\tau}-LSN_{\tau-1}\right]+cte,&\end{aligned}\tag{35}$$

where cte is a constant. We are interested in the conditional expectation of that quantity, and so

$$\begin{aligned}&\mathbb{E}\left[\log\frac{LSO_{\tau}-LS_{\tau}}{LS_{\tau}-LSN_{\tau}}-\log\frac{LSO_{\tau-1}-LS_{\tau-1}}{LS_{\tau-1}-LSN_{\tau-1}}|\varepsilon^{s}_{\omega}\right]\\ &=cte+\left(\frac{1}{\overline{LS}-\overline{LSN}}+\frac{1}{\overline{LSO}-\overline{LS}}\right)\mathbb{E}\left[LS_{\tau-1}-LS_{\tau}|\varepsilon^{s}_{\omega}\right],\end{aligned}$$

where we also use Assumption 2 to include terms in the constant. We combine this last expression with (19) and (33) to obtain

$$\left(\frac{1}{\overline{LS}-\overline{LSN}}+\frac{1}{\overline{LSO}-\overline{LS}}\right)\mathbb{E}\left[LS_{\tau-1}-LS_{\tau}|\varepsilon^{s}_{\omega}\right]=cte+\mathbb{E}\left[\frac{\sigma_{\omega}-1}{1-\bar{s}}\right]\frac{1}{\mathcal{L}}\mathbb{E}\left[\frac{x\left(\tau,\omega\right)}{l\left(\tau,\omega\right)}|\varepsilon^{s}_{\omega}\right].$$

Q.E.D.

Proof of Proposition 1

Before we begin, we establish a preliminary technical result on the equivalence of two related statistical models.

Suppose that the data-generating process is

$$
\begin{aligned}
Z &= X\beta_A + v_A \\
X &= M_Y V\alpha_A + u_A,
\end{aligned}
$$

where α_A and β_A are coefficient vectors, v_A and u_A are error term vectors, X and V are predetermined matrices, and where

$$M_Y = I - P_Y = I - Y\left(Y'Y\right)^{-1}Y'$$

is the orthogonal (linear) projection matrix (also known as annihilator matrix) that returns residual from projecting any conformable matrix on the subspace generated by Y (a predetermined variable). P_Y is the (linear) projection matrix that returns predicted values (also known as prediction matrix). We will refer to this system of equations as model A.

Consider an alternative model (model B) that follows the equations

$$
\begin{aligned}
Z &= X\beta_B + Y\gamma_B + v_B \\
X &= V\alpha_B + Y\delta_B + u_B,
\end{aligned}
$$

where α_B, β_B, δ_B and γ_B are coefficient vectors and v_B and u_B are error term vectors that are orthogonal to Yin the sense that $M_Y v_B = M_Y u_B = 0$. Note that the above definition implies that the error terms are related as follows:

$$X\beta_A + v_A = X\beta_B + Y\gamma_B + v_B$$

and

$$V\alpha_B + Y\delta_B + u_B = M_Y V\alpha_A + u_A,$$

which after multiplying both sides by the orthogonal projection matrix M_Y implies

$$M_Y V\alpha_B + M_Y u_B = M_Y V\alpha_A + M_Y u_A. \tag{36}$$

The parameters for model A and model B are identical and their estimates have identical properties, as stated in the lemma below.

Lemma 6. *The two-stage least squares estimators of model A and model B yield the same point estimates for β_A and β_B and for α_A and α_B, and these estimators have the same variance.*

The proof of this fact is a straightforward corollary of a standard result in econometrics: the Frisch–Waugh–Lovell theorem. We give this standard, technical proof below.

The proof of the Proposition 1 now trivially follows from Lemma (4) and Lemma (5) and the basic properties of a conditional expectations operator, i.e., the fact that for any random variable X and Y (with bounded expectation) we have $Y = \mathbb{E}[Y|X] + \varepsilon$, we know 1) $\mathbb{E}[\varepsilon|X] = 0$, 2) $\mathbb{E}[\varepsilon] = 0$, 3) for any function $h(X)$ such that $\mathbb{E}[|h(X)\varepsilon|] < \infty$, $\mathbb{E}[h(X)\varepsilon] = 0$. It is clear that the first stage boils down to model A. Under the independence assumption the terms s_i, s_c are random variables that can be estimated by a fixed effect regression of the form $s = FE_c + FE_i + \varepsilon$ (see Wooldridge (2002), section 10.2.1). This is equivalent to estimating model B as stated in the proposition by the above lemma. We give the detailed proof below. Q.E.D.

Proof of Lemma 6

We first focus on model B. We can multiply both equations by M_Y to find

$$\begin{aligned} M_Y Z &= M_Y X \beta_B + M_Y v_B \\ M_Y X &= M_Y V \alpha_B + M_Y u_B, \end{aligned}$$

where we have used the key property of orthogonal projection matrix that $M_Y Y = 0$ and $M_Y v_B = 0$, $M_Y u_B = u_B$, since v_B and u_B are both orthogonal to Y. From the Frisch–Waugh–Lovell theorem we know that the estimate of the coefficients using these equations is the same as the estimate of the coefficients of the original model B.

In the first-stage estimation, we can use the standard equation for ordinary least squares coefficients to write

$$\begin{aligned} \hat{\alpha}_B &= \left[(M_Y V)' M_Y V\right]^{-1} (M_Y V)' M_Y X \\ &= \left[V' M_Y V\right]^{-1} V' M_Y X \end{aligned}$$

since M_Y is idempotent and symmetric. The predicted value is of the dependent variable is therefore

$$\begin{aligned} \widehat{M_Y X} &= M_Y V \hat{\alpha}_B \\ &= M_Y V \left[V' M_Y V\right]^{-1} V' M_Y X \\ &= P_{M_Y V} X \end{aligned}$$

where $P_{M_Y V} = M_Y V \left[V' M_Y V\right]^{-1} V' M_Y$ is the projection matrix on the subspace generated by

$M_y V$. As a result, we can write the second stage estimation as

$$M_Y Z = P_{M_Y V} X \beta_B + M_Y v_B$$

and, using the standard ordinary least squares formula, the estimate of β_B is

$$\hat{\beta}_B = (X' P_{M_Y V} X)^{-1} X' P_{M_Y V} M_Y Z.$$

We now turn to model A. In the first stage, we can compute the estimate of α_A as

$$\hat{\alpha}_A = (V' M_Y V)^{-1} (M_Y V)' X$$

so that the predicted value of $\hat{X}$ is

$$\hat{X} = P_{M_Y V} X.$$

The second stage becomes,

$$\begin{aligned} Z &= \hat{X} \beta_A + v_A \\ &= P_{M_Y V} X \beta_A + v_A \end{aligned}$$

and so the estimate of β_A is

$$\begin{aligned} \hat{\beta}_A &= \left((P_{M_Y V} X)' P_{M_Y V} X \right)^{-1} (P_{M_Y V} X)' Z \\ &= (X' P_{M_Y V} X)^{-1} X' P_{M_Y V} Z. \end{aligned}$$

where we have used the fact that $P_{M_Y V}$ is idempotent and symmetric. It follows that the two estimators $\hat{\beta}_A$and $\hat{\beta}_B$ are equivalent if

$$P_{M_Y V} M_Y = P_{M_Y V}.$$

We can compute the left-hand side as

$$\begin{aligned} P_{M_Y V} M_Y &= M_Y V \left((M_Y V)' M_Y V \right)^{-1} (M_Y V)' M_Y \\ &= M_Y V \left((M_Y V)' M_Y V \right)^{-1} V' M_Y \\ &= P_{M_Y V}, \end{aligned}$$

where we have used again the fact that M_Y is idempotent. This completes the proof that $\hat{\beta}_A = \hat{\beta}_B$.

Note also that the expression for $\hat{\alpha}_A$ and $\hat{\alpha}_B$ are identical and so $\hat{\alpha}_A = \hat{\alpha}_B$. Since we have shown that the residuals are exactly the same between the two models the variance of coefficients must also be identical. Q.E.D.

Detailed proof of Proposition 1

We now turn to the proof of Proposition 1. Let $s_\omega := s(\omega,\tau)$, $\varepsilon^s_\omega := \varepsilon^s(\omega,\tau)$, which is without loss under Assumption 2. The goal is to show that the exclusion restriction is satisfied.

From the basic properties of the conditional expectations operator, we know that[48]

$$\begin{aligned} LS(\omega,\tau) - LS(\omega,\tau-1) &= \mathbb{E}\left[LS(\omega,\tau) - LS(\omega,\tau-1)\,|\varepsilon^s_\omega\right] + \varepsilon_1(\omega), \\ \frac{x(\omega,\tau)}{l(\omega,\tau)} &= \mathbb{E}\left[\frac{x(\omega,\tau)}{l(\omega,\tau)}|\varepsilon^s_\omega\right] + \varepsilon_2(\omega), \end{aligned}$$

where $\mathbb{E}\left[\varepsilon_1(\omega)\,|\varepsilon^s_\omega\right] = 0$ and $\mathbb{E}\left[\varepsilon_2(\omega)\,|\varepsilon^s_\omega\right] = 0$. Combining with (19) and (20) we write

$$LS(\omega,\tau) - LS(\omega,\tau-1) = cte + \mathcal{B}\frac{x(\omega,\tau)}{l(\omega,\tau)} + \underbrace{\varepsilon_1(\omega) - \mathcal{B}\varepsilon_2(\omega)}_{\varepsilon_3(\omega)}, \tag{37}$$

$$\frac{x(\omega,\tau)}{l(\omega,\tau)} = cte + \mathcal{L}(s_\omega - s_i - s_c) + \varepsilon_2(\omega). \tag{38}$$

Finally, note that

$$\begin{aligned} \mathbb{E}\left[(s - s_i - s_c)\,\varepsilon_3(\omega)\right] &= \mathbb{E}\left[\varepsilon^s_\omega \varepsilon_3(\omega)\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\varepsilon^s_\omega \varepsilon_3(\omega)\,|\varepsilon^s_\omega\right]\right] \\ &= \mathbb{E}\left[\varepsilon^s_\omega \mathbb{E}\left[\varepsilon_3(\omega)\,|\varepsilon^s_\omega\right]\right] \\ &= 0, \end{aligned}$$

and so ε^s_ω satisfies the exclusion restriction necessary to obtain an unbiased and consistent estimate of $\mathcal{B}$. Finally, we want to show that $\varepsilon_2(\omega)$ is uncorrelated with $(s - s_i - s_c)$. The steps here are identical as for $\varepsilon_3(\omega)$ above:

$$\begin{aligned} \mathbb{E}\left[(s - s_i - s_c)\,\varepsilon_2(\omega)\right] &= \mathbb{E}\left[\varepsilon^s_\omega \varepsilon_2(\omega)\right] \\ &= \mathbb{E}\left[\varepsilon^s_\omega \mathbb{E}\left[\varepsilon_2(\omega)\,|\varepsilon^s_{ci}\right]\right] \\ &= 0. \end{aligned}$$

[48]Let X and Y be random variables such that $\mathbb{E}[Y] < \infty$. We use the following properties of $\varepsilon := Y - \mathbb{E}[Y|X]$ throughout: 1) $\mathbb{E}[\varepsilon|X] = 0$, 2) $\mathbb{E}[\varepsilon] = 0$, 3) for any function $h(X)$ such that $\mathbb{E}[|h(X)\varepsilon|] < \infty$, $\mathbb{E}[h(X)\varepsilon] = 0$.

Finally, note that the system (37)–(38) fits in the description of model A in Lemma 6 (the expression $s - s_i - s_c$ corresponds to the residual after projecting on industry and city fixed effects, corresponding to Y in the lemma). Under the independence assumption the terms s_i, s_c are random variables that can be estimated by a fixed effect regression of the form $s = FE_c + FE_i + \varepsilon$ (see discussion in Wooldridge (2002), section 10.2.1). We therefore know that we can instead estimate $\mathcal{B}$ and $\mathcal{L}$ using model B, as in the statement of the proposition.

E Robustness to correlated pre-existing subsidies

In this section, we generalize the theoretical results of Section 2 to allow for a pre-existing subsidy for automation investment in the period $\tau - 1$, which would have violated Assumption 2. More specifically, we assume that the part of subsidies $s(\tau, \omega)$ and $s(\tau - 1, \omega)$ that are orthogonal to the industry and city variables are related as follows:

$$\varepsilon^s (\tau - 1, \omega) = \rho^s_\omega \varepsilon^s (\tau, \omega) + v^s_\omega, \tag{39}$$

where ρ^s_ω and v^s_ω are i.i.d. shocks with mean 0. The results that are affected by this change are Lemma 4, Lemma 5 and Proposition 1. We give the modified proofs below. The resulting statement of the proposition is identical. The only difference is that the constant $\mathcal{B}$ that is carried around is given by

$$\mathcal{B} = -\frac{1}{1 - \bar{s}} \frac{1}{\mathcal{L}} \left(\frac{1}{\overline{LS} - \overline{LSN}} + \frac{1}{\overline{LSO} - \overline{LS}} \right)^{-1} \mathbb{E}\left[(\sigma_\omega - 1)(1 - \rho^s_\omega) \right], \tag{40}$$

instead of by (21).

Implications for estimating the average elasticity $\mathbb{E}[\sigma_\omega]$

We now show that the above modification implies that, when $0 < \mathbb{E}[\rho^s_\omega]$, our estimated elasticity is biased toward a smaller $\mathbb{E}[\sigma_\omega]$ and, as a result, against finding a large effect of automation on the labor share. To see this, note that we can find $\mathbb{E}[\sigma_\omega]$ by using the estimated coefficients $\hat{\mathcal{B}}$ and $\hat{\mathcal{L}}$. Namely, from (40) we can write

$$\mathbb{E}[\sigma_\omega - 1] = -\hat{\mathcal{L}}\hat{\mathcal{B}} \frac{1 - \bar{s}}{\mathbb{E}[1 - \rho^s_\omega]} \left(\frac{1}{\overline{LS} - \overline{LSN}} + \frac{1}{\overline{LSO} - \overline{LS}} \right) \tag{41}$$

where we have used the fact that ρ^s_ω is independent of other random variables. It is clear that the right-hand side of (41) is an increasing function of $\mathbb{E}[\rho^s_\omega]$ (the product $\hat{\mathcal{L}}\hat{\mathcal{B}}$ is always negative in our estimates and $1 - \bar{s}$, $\overline{LS} - \overline{LSN}$ and $\overline{LSO} - \overline{LS}$ are positive by the restrictions of the model). As a result, by assuming that $\mathbb{E}\left[\rho^s_\omega\right] = 0$ as in our benchmark exercises, we would be biasing our estimate toward a smaller $\mathbb{E}\left[\sigma_\omega\right]$.

In this section, we assume that the subsidy are given by

$$s(\tau,\omega) = s_i(\tau) + s_c(\tau) + \varepsilon^s(\tau,\omega)$$
$$s(\tau-1,\omega) = s_i(\tau-1) + s_c(\tau-1) + \varepsilon^s(\tau-1,\omega)$$

where the residuals are related as

$$\varepsilon^s(\tau-1,\omega) = \rho^s_\omega \varepsilon^s(\tau,\omega) + v^s_\omega, \tag{39}$$

where ρ^s_ω and v^s_ω are i.i.d. shocks with mean 0.

Modified Lemma 4

Lemma 7. *The automation investment decision is related to the subsidy residual through the equation*

$$\mathbb{E}\left[\frac{x(\omega,\tau)}{l(\omega,\tau)}|\varepsilon^s_{ci}\right] \approx \mathcal{L}\varepsilon^s_{ci} + cte, \tag{42}$$

where $\mathcal{L}$ and cte are some constants.

Proof. From (29), we can write

$$\frac{x(\tau,\omega)}{l(\tau,\omega)} = p_m(\tau)\exp\left(\Theta(\tau,\omega)\right)(1-s_\omega)^{-\sigma_\omega} - (1-\delta^m)\, p_m(\tau-1)\exp\left(\Theta(\tau-1,\omega)\right)(1-s_{\omega,-1})^{-\sigma_\omega}\frac{l(\tau-1,\omega)}{l(\tau,\omega)}.$$

We begin by taking a linear approximation of $\frac{x(\omega,\tau)}{l(\omega,\tau)}$ around the subsidies that the firm faces to write

$$\frac{x(\omega,\tau)}{l(\omega,\tau)} \approx \mathcal{N}(\tau-1,\tau,\omega) + \mathcal{H}_\tau(\tau-1,\tau,\omega)\, s_\omega + \mathcal{H}_{\tau-1}(\tau-1,\tau,\omega)\, s_{\omega,-1}$$

where $\mathcal{N}(\tau-1,\tau,\omega)$, $\mathcal{H}_\tau(\tau-1,\tau,\omega)$ and $\mathcal{H}_{\tau-1}(\tau-1,\tau,\omega)$ do not depend on the subsidies (they can depend however on the linearization point $\bar{s}$). Using (17) and (39) we can rewrite that equation as

$$\begin{aligned}
\frac{x(\omega,\tau)}{l(\omega,\tau)} &\approx \mathcal{N}(\tau-1,\tau,\omega) + \mathcal{H}_\tau(\tau-1,\tau,\omega)\left(s_i(\tau) + s_c(\tau) + \varepsilon^s(\tau,\omega)\right) \\
&\quad + \mathcal{H}_{\tau-1}(\tau-1,\tau,\omega)\left(s_i(\tau-1) + s_c(\tau-1) + \rho^s_\omega\varepsilon^s(\tau,\omega) + v^s_\omega\right) \\
&\approx \mathcal{N}(\tau-1,\tau,\omega) + \mathcal{H}_\tau(\tau-1,\tau,\omega)\left(s_i(\tau) + s_c(\tau)\right) + \mathcal{H}_{\tau-1}(\tau-1,\tau,\omega)\left(s_i(\tau-1) + s_c(\tau-1) + v^s_\omega\right) \\
&\quad + \left(\mathcal{H}_\tau(\tau-1,\tau,\omega) + \mathcal{H}_{\tau-1}(\tau-1,\tau,\omega)\,\rho^s_\omega\right)\varepsilon^s(\tau,\omega)
\end{aligned}$$

Using Assumption 2 and taking the conditional expectation on both sides we find

$$\mathbb{E}\left[\frac{x\left(\omega,\tau\right)}{l\left(\omega,\tau\right)}|\varepsilon_{ci}^{s}\right]=\mathcal{L}\varepsilon_{ci}^{s}+cte$$

where

$$\mathcal{L}=\mathbb{E}\left[\left(\mathcal{H}_{ci}\left(\tau\right)+\mathcal{H}_{ci}\left(\tau-1\right)\rho_{\omega}^{s}\right)\right],$$

which completes the proof. □

Modified Lemma 5

Lemma 8. *The change in the labor share is related to automation investment through equation*

$$\mathbb{E}\left[LS\left(\omega,\tau\right)-LS\left(\omega,\tau-1\right)|\varepsilon_{ci}^{s}\right]\approx\mathcal{B}\mathbb{E}\left[\frac{x\left(\omega,\tau\right)}{l\left(\omega,\tau\right)}|\varepsilon_{ci}^{s}\right]+cte \tag{43}$$

where cte is a constant and where $\mathcal{B}$ is the constant given by (40), with $\bar{s}$ the average subsidy, $\overline{LS}$ the average labor share, and $\overline{LSO}$ and $\overline{LSN}$ the averages of (4) and (15), respectively.

Proof. Recall 32 that we derived in the proof of Lemma 5. Taking the difference of that equation between τ and $\tau-1$ yields

$$\begin{aligned}\log\frac{LSO_{\tau}-LS_{\tau}}{LS_{\tau}-LSN_{\tau}}-\log\frac{LSO_{\tau-1}-LS_{\tau-1}}{LS_{\tau-1}-LSN_{\tau-1}}&=\Psi\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{\sigma_{\omega}}c_{i}\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{1-\bar{s}}s\left(\tau,\omega\right)\\&\quad-\Psi\left(\tau-1,\omega\right)-\frac{\sigma_{\omega}-1}{\sigma_{\omega}}c_{i}\left(\tau-1,\omega\right)-\frac{\sigma_{\omega}-1}{1-\bar{s}}s\left(\tau-1,\omega\right),\end{aligned}$$

or, using (17) and (39),

$$\begin{aligned}&\log\frac{LSO_{\tau}-LS_{\tau}}{LS_{\tau}-LSN_{\tau}}-\log\frac{LSO_{\tau-1}-LS_{\tau-1}}{LS_{\tau-1}-LSN_{\tau-1}}\\&=\Psi\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{\sigma_{\omega}}c_{i}\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{1-\bar{s}}\left(s_{i}\left(\tau\right)+s_{c}\left(\tau\right)+\varepsilon^{s}\left(\tau,\omega\right)\right)\\&\quad-\Psi\left(\tau-1,\omega\right)-\frac{\sigma_{\omega}-1}{\sigma_{\omega}}c_{i}\left(\tau-1,\omega\right)-\frac{\sigma_{\omega}-1}{1-\bar{s}}\left(s_{i}\left(\tau-1\right)+s_{c}\left(\tau-1\right)+\rho_{\omega}^{s}\varepsilon^{s}\left(\tau,\omega\right)+v_{\omega}^{s}\right)\\&=\Psi\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{\sigma_{\omega}}c_{i}\left(\tau,\omega\right)+\frac{\sigma_{\omega}-1}{1-\bar{s}}\left(s_{i}\left(\tau\right)+s_{c}\left(\tau\right)\right)+\frac{\sigma_{\omega}-1}{1-\bar{s}}\left(1-\rho_{\omega}^{s}\right)\varepsilon^{s}\left(\tau,\omega\right)\\&\quad-\Psi\left(\tau-1,\omega\right)-\frac{\sigma_{\omega}-1}{\sigma_{\omega}}c_{i}\left(\tau-1,\omega\right)-\frac{\sigma_{\omega}-1}{1-\bar{s}}\left(s_{i}\left(\tau-1\right)+s_{c}\left(\tau-1\right)+v_{\omega}^{s}\right).\end{aligned}$$

Taking the conditional expectation on both sides and using Assumption 2, the rest of the proof follows the same steps as in Lemma 5, the only difference being the $1-\rho_{\omega}^{s}$ in the above expression. As a result, the statement of the result is unchanged except that the constant $\mathcal{B}$ is different. □

Modified Proposition 1

Proposition 2. *The coefficients $\mathcal{B}$ and $\mathcal{L}$ can be consistently estimated from two-stage least squares regression of the form:*

$$LS(\tau,\omega) - LS(\tau-1,\omega) \approx cte + \mathcal{B}\frac{x(\tau,\omega)}{l(\tau,\omega)} + FE_i + FE_c + e(\omega), \tag{44}$$

$$\frac{x(\tau,\omega)}{l(\tau,\omega)} \approx cte + \mathcal{L}s_\omega + FE_i + FE_c + u(\omega), \tag{45}$$

where e, u are error terms, $\mathcal{B}$ is the constant (40) and FE_i, FE_c are a set of industry and city fixed effects, respectively.

Proof. The only difference is that the constant $\mathcal{B}$ that is carried around is given by (40) instead of (21). □

F Details of the calibration

This appendix collects the mechanics of the calibration summarized in Section 4.1: the structure of the simulated economy, the moments and their definitions, the parameters set outside the estimation, and the diagnostics that the model is not asked to match.

The simulated economy

We work with two periods, a pre-subsidy period $\tau - 1$ that we map to 2015 and a post-subsidy period τ that we map to 2017, and relate model differences between τ and $\tau - 1$ to data differences between the two years. The simulated economy reproduces the city-industry skeleton of the survey: 44 cities, 23 industries, and the 122 city-industry groupings that contain at least one automating firm. Each of the 122 groupings is populated with the same fixed number of firms, 28 in the calibration, of which the three with the highest period-τ demand automate. Using this many firms per grouping keeps the calibration moments and equilibrium prices stable rather than driven by the handful of firms the survey samples in a cell; adding more changes the moments negligibly. We compute the calibration moments on this full simulated economy. For the validation of Section 4.3 we scale both counts up by a factor of four (112 firms per grouping, twelve automating) to form a large-economy "population," from which we draw subsamples whose size and distribution across groupings match the survey's 177 automating firms.

Setting a firm's automation capability to $a_\omega < 1$ does not by itself make the firm automate, since a firm facing weak demand may choose not to invest. To reproduce the observed pattern of adoption, we designate as automating, within each grouping, the firms with the highest period-τ demand, those large enough to find investment in automation worthwhile. This rule is the reduced form of a profit-maximizing adoption decision: larger firms adopt. It also generates exactly the dependence between automation and firm-level demand that the subsidy instrument is designed to purge.

Means of the technology parameters

The choice of $\overline{LSO}$ pins the common intensities $\alpha = \gamma$ and the mean of η through the labor-share identity (4). Using $\overline{LSOP} = \frac{\theta-1}{\theta}(1-\overline{\eta})(1-\alpha)$,

$$\alpha = \gamma = 1 - \frac{\theta}{\theta-1}\overline{LSO}, \qquad \overline{\eta} = 1 - \frac{\overline{LSOP}}{\overline{LSO}}.$$

Automation lowers the labor share only through production activities, so $\overline{LS}_\tau - \overline{LSO} = \overline{LSP}_\tau - \overline{LSOP}$. Combining this with the display above and the automating-firm averages $\overline{LS}_\tau = .456$ and $\overline{LSP}_\tau = .308$ (Table 2) gives $\overline{\eta} \approx .25$ and $\alpha = \gamma \approx .20$ at $\overline{LSO} = .60$.

Shocks and trend growth

Firm size and its evolution come from the demand shifter. The initial log demand is drawn i.i.d., and the period-τ value follows a random walk:

$$\log d(\omega, \tau-1) \sim \mathcal{N}\big(0, \sigma_{d(\tau-1)}\big), \qquad \log d(\omega, \tau) \sim \mathcal{N}\big(\log d(\omega, \tau-1), \sigma_{d(\tau)}\big).$$

The first standard deviation controls the dispersion of firm size; the second controls the dispersion of firm growth. Because demand d and productivity A enter the firm's problem in the same way, we use d for idiosyncratic size and growth and reserve A for aggregate trends: productivity grows at a common rate so that the model matches manufacturing wage growth of 12.5 percent. Since technological progress here is not labor-augmenting and aggregate labor supply is fixed, a rise in productivity raises wages by more than productivity itself, which is what spurs automation.

User cost of capital and the price of automation

We compute user costs from (5). We take the real cost of funds to be 6 percent annually, the emerging-market weighted average cost of capital for 2016 net of US inflation, and set annual depreciation at 5 percent for structures (support capital k_s) and 10 percent for equipment k_e and automation capital m, consistent with a useful life of about ten years for industrial robots and digitally controlled machinery. These rates are common across firms and time and are compounded to three-year equivalents; they matter little for the results.

The price of automation p_m and the mean capability $\overline{a}$ play equivalent roles, so we set $\overline{a} = .5$ with no dispersion and let p_m carry the information. The initial price $p_m(\tau - 1)$ is chosen so that the model reproduces the 2015 labor share of automating firms: it determines how much automation capital these firms had already accumulated by 2015. The change in p_m is pinned by the growth of the automation stock. Industrial-robot density in Chinese manufacturing rose by 98 percent between 2015 and 2017 (International Federation of Robotics), so we require the value added-weighted automation stock to roughly double. In the baseline this implies a real decline in the automation price of $\Delta_{\%} p_m = -2.1$ percent (-3.8 percent in the lower-elasticity calibration). The modest size of this decline is consistent with the external evidence: industrial-robot prices were roughly flat in dollar terms over this period, and the renminbi depreciated, so the real decline reflects quality improvement rather than a fall in the transacted price.

Subsidies enter at the city-industry level: only the groupings that receive positive subsidies in the data receive them in the model, at the rates observed in the survey (winsorized at the 90th percentile and capped at 50 percent). The mean subsidy among subsidized firms is about 0.228, the value used in the elasticity back-out of Section 3. In the validation of Section 4.3, we discipline the simulated subsidy level to this same back-out point so that the recovery exercise is the exact analog of the empirical one.

Identification of $\overline{LSO}$

We adopt $\overline{LSO} = .60$ from the preferred specification of Section 3, but the data also speak to it directly through the growth of automating firms. A higher $\overline{LSO}$ raises the automation-free benchmark from which the labor share must fall to reach its observed 2015 level, so the model infers a larger initial automation stock. Because the automation stock is then required to double, a larger initial stock means a larger absolute increase, a steeper fall in the labor share, and a sharper rise in value added. At $\overline{LSO} = .66$ the implied growth in the value added of automating firms overshoots what we see in the data, which is why the lower value is preferred. The unit-elasticity case, in which the labor share is invariant to the price of automation, is rejected even more sharply: it would require an implausible explosion in the value added of automating firms.[49]

[49] With markups of 33 percent, $\overline{LSO}$ cannot exceed about .75, so the textbook value of .66 is already close to the upper bound consistent with the assumed markup.

Diagnostics not targeted

Table 14 reports moments that the calibration does not target. They mark the dimensions along which a uniform-markup model without a firm-level productivity premium reaches its limits. The model overstates the value-added share of automating firms and understates the dispersion of their size, because it makes automating firms large through demand alone. It overstates the value-added growth of automating firms, the counterpart of the same force. And it overstates the aggregate labor share, because the non-automating firms that produce most of value added are held at the automation-free benchmark $\overline{LSO} = .60$ rather than at their lower measured labor share. The model is designed to capture automating firms, and these diagnostics show that it does so at the cost of the cross-sectional size distribution and the level of the aggregate labor share. Introducing heterogeneity in markups is the natural refinement and is left for future work.

Diagnostic (not targeted)	Data	Model
Value-added share of automating firms	.25	.31
c.v. of employment, automating firms	1.01	.64
Value-added growth, automating firms	1.32	1.43
Aggregate labor share, all firms	.50	.57

Table 14: Untargeted diagnostics of the calibrated model

Notes: Moments the calibration does not target. The aggregate labor share is the economy-wide measured value ($\approx .50$), distinct from the value added-weighted mean over the calibration sample reported for all firms in Table 7 (.477).

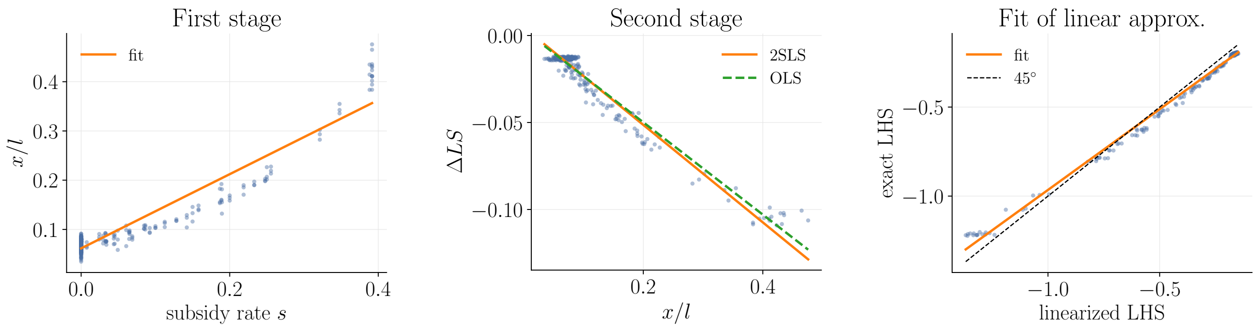
G Details of the validation

This appendix supports the validation of Section 4.3 with two elements kept out of the main text: a homogeneous benchmark, in which the estimator's linear approximation is exact, and an exact version of the estimator. Together they locate the small wedge between the recovered and the true elasticity in the linear approximation and confirm that the identification itself is sound.

A homogeneous benchmark

Figure 3 runs the estimator on a near-homogeneous economy, with the dispersion in η and demand turned off. Two features stand out. First, in the second stage (panel b) the ordinary-least-squares and instrumental-variable slopes coincide: with firms alike, automation is not confounded with unobserved firm characteristics. In the calibrated economy this confounding is present but mild, because a uniform-markup model generates only limited cross-firm dispersion;

the importance of the instrument is therefore an empirical matter, established on the data in Section 3, not a feature the model is meant to reproduce. Second, and the reason we display this economy, panel (c) plots the linear approximation the estimator relies on (the linearized left-hand side of (35)) against the exact nonlinear object: under homogeneity the two lie on the 45-degree line, so the approximation is exact and the estimator recovers the elasticity without error.



(a) First stage

(b) Second stage

(c) Fit of the linear approximation

Figure 3: Estimation on data generated by the near-homogeneous model. The OLS and IV slopes in panel (b) coincide; panel (c) shows the linear approximation lying on the 45-degree line.

The linear approximation and the exact estimator

Heterogeneity curves the relationship the estimator linearizes. Figure 4 repeats the approximation-fit plot for the calibrated economy: the points now scatter around the 45-degree line, and this curvature is the source of the roughly eight percent under-recovery reported in the main text.

To confirm that this curvature, and not the identification, drives the wedge, we compute an *exact* estimator that uses the nonlinear log-deviation in place of its linear approximation. This estimator is a diagnostic only: it requires each firm's automation-free labor share $LSO(\omega)$, a counterfactual that the data do not contain, which is exactly why the empirical strategy linearizes it away. Run on the model, the exact estimator recovers 3.12 in the population and 3.05 in the 177-firm sample, against a truth of 3.04. It recovers the elasticity almost exactly, in the population and the small sample alike. Identification is therefore sound: the subsidy instrument isolates the variation in automation prices, and the general-equilibrium response of prices to subsidies, differenced out by the city and industry fixed effects, does not bias the estimate. The entire wedge in the linearized estimator of Section 4.3 is the price of the approximation.

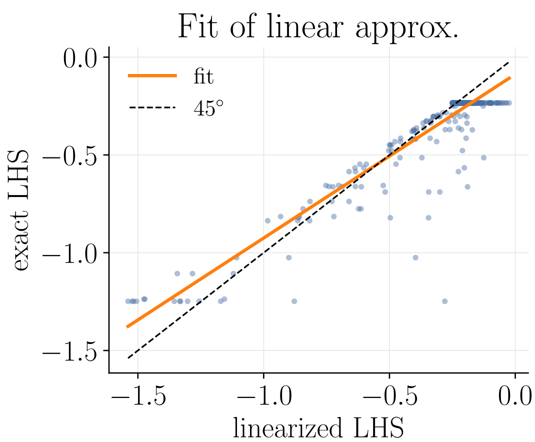


Figure 4: Fit of the linear approximation in the calibrated model: the linearized left-hand side of (35) against the exact nonlinear log-deviation. The scatter around the 45-degree line is the curvature the approximation neglects.